

claude-windows / c6028e72-ebfd-4218-89a8-c4d7dd23be96

Metadata

```
- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\c6028e72-ebfd-4218-89a8-c4d7dd23be96\subagents\workflows\wf_49aab888-e10\agent-a3aa151cba9f04970.jsonl`
- SHA-256: `df983d0ee6af48876776c0f2c42c4f7e112740243d354a3415e59eeffa09ec23`
- Source modified: `2026-06-08T13:17:39+00:00`
- Imported at: `2026-07-05T16:48:23+00:00`
- project: `wf_49aab888-e10`
- session_id: `c6028e72-ebfd-4218-89a8-c4d7dd23be96`
```

Transcript

1. user (2026-06-08T13:13:45.328Z)

Adversarially VERIFY the load-bearing claims in these research findings, focus: "exact licenses of every named base model (Apache/MIT vs Gemma-3-style/community/NC) + video-capability claims". Use WebSearch + WebFetch to check each against primary sources (official pricing pages, model license files / HuggingFace cards, provider DPAs). Default to 'uncertain'/'outdated' if you cannot confirm from a primary source. Flag anything that would change a recommendation. Be specific.

RESEARCH FINDINGS:

```
[
  {
    "dimension": "Training compute platforms for an owned video-QA (rally temporal segmentation) model",
    "summary": "A LoRA/QLoRA fine-tune of an Apache-2.0 7B VLM on our 5k-20k clip-QA pairs is a small job: under 24h on one A100 80GB, $10-50 total. So the deciding factor is privacy terms and operational ergonomics, not GPU dollar-per-hour. Modal has the strongest written no-access contract; Vertex AI is the only strong-privacy option with an India region. The RTX 2070 Super 8GB is dev/inference-only, too small for real 7B-VLM training.",
    "options": [
      {
        "name": "Self-hosted RTX 2070 Super 8GB + WSL2",
        "what": "Local single-GPU fine-tuning/inference via ms-swift/Unsloth/QLoRA. Zero marginal cost, zero data egress.",
        "pros": [
          "Maximal privacy: data never leaves the machine - the local-first moat",
          "Zero cost; ideal free dev/CI smoke-test box and 4-bit 7B inference",
          "No lock-in, no DPA, no billing surprises"
        ],
        "cons": [
          "8GB too small for a real 7B-VLM QLoRA run (needs ~16-24GB with vision/video tokens; 4-bit 7B text inference alone ~7.1GB)",
          "Turing lacks bf16/modern FlashAttention kernels",
          "Single GPU; unsuitable for Option B end-to-end multimodal training"
        ],
        "cost_note": "$0 marginal (owned). Dev/CI/inference only.",
        "license": "N/A (own hardware)",
        "privacy_note": "BEST possible - data never leaves device.",
        "fit_for_us": "medium"
      },
      {
        "name": "Modal (serverless GPU) - RECOMMENDED PRIMARY",
        "what": "Serverless Python-native compute, per-second billing, scale-to-zero, A100/H100. Has a Training product.",
        "pros": [
          "Strongest privacy contract: Modal 'will never access your source code, the
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inputs/outputs to your Functions, or any data you store'; inputs/outputs deleted within 7 days;
SOC2 Type II; HIPAA via BAA",
    "Per-second billing + scale-to-zero = no idle cost, perfect for retrain-on-each-
golden-batch cadence",
    "Python/recipe-driven - fits a one-command agent-runnable harness and the future
ComputeBackend abstraction",
    "Asia-Pacific regions (Singapore, Seoul); $30/mo free credits"
],
"cons": [
    "No India asia-south1 region (Singapore closest) - fine for batch training but not
India residency-at-rest",
    "A100 ~$3.72/hr, H100 ~$4.29/hr - pricier per-hour than RunPod/Vast",
    "Regional multipliers 1.25x-2.5x for non-US"
],
"cost_note": "A100 80GB ~$3.72/hr, H100 ~$4.29/hr per-second. Full LoRA fine-tune
~$10-$50. $30/mo free credits.",
"license": "Commercial cloud; you keep all weights/data. Qwen2.5-VL-7B Apache-2.0 base
clean.",
"privacy_note": "STRONGEST written no-access/short-retention terms found. Meets the hard
requirement.",
"fit_for_us": "strong"
},
{
    "name": "RunPod (Secure Cloud tier) - RECOMMENDED FALLBACK",
    "what": "Rented GPU pods + serverless. Secure Cloud (vetted DCs) vs Community Cloud
(unvetted hosts).",
    "pros": [
        "Cheap A100 80GB ~$1.19-$1.29/hr, H100 ~$2.65/hr (spot ~$1.19) - best value for
longer/multi-GPU runs",
        "Secure Cloud: SOC2 Type II, ISO 27001, GDPR, HIPAA; AES-256 at rest; ToS bars hosts
inspecting pod data",
        "Per-tenant container isolation; instant data destruction on delete; serverless +
checkpointing; has a DPA"
    ],
    "cons": [
        "Community Cloud tier unvetted peer-to-peer - MUST restrict corpus to Secure Cloud
only",
        "No India/Asia datacenter - no India residency",
        "Less one-command-reproducible than Modal; USD billing"
    ],
    "cost_note": "A100 80GB ~$1.19-$1.29/hr (Secure), H100 ~$2.65/hr on-demand / ~$1.19
spot. Our fine-tune ~$3-$15.",
    "license": "Commercial cloud; you keep weights/data.",
    "privacy_note": "STRONG on Secure Cloud. MUST avoid Community Cloud for sensitive
data.",
    "fit_for_us": "strong"
},
{
    "name": "Google Vertex AI / GCP Compute Engine GPUs - KEEP ON BENCH",
    "what": "Managed Vertex custom-training jobs or raw Compute Engine GPU VMs. The platform
the user named.",
    "pros": [
        "Written: 'Customer data is never used by Google. We do not use customer data to train
our models, or models used by others, without permission'",
        "India region asia-south1 (Mumbai) for Compute Engine GPUs - genuine India residency-
at-rest, unique among strong options",
        "Mature multi-GPU/checkpointing/VPC/CMEK; strong ecosystem maps to future
Storage/Compute backends"
    ],
    "cons": [
        "Most expensive + most overhead: Vertex mgmt fee (A100 ~$3.37/hr all-in, H100
~$11.27/hr); easy to overspend on always-on resources",
        "Heavyweight IAM/quota/capacity friction; slowest path to a reproducible harness",
        "gen-AI residency region list excluding asia-south1 applies to MANAGED gen-AI APIs,
not custom training on your own weights (asia-south1 GPUs ARE usable)"
    ]
}

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    ],
    "cost_note": "A100 ~$3.37/hr all-in, H100 ~$11.27/hr; L4/T4 cheap. Highest per-hour +
overhead.",
    "license": "Commercial cloud; you keep weights/data.",
    "privacy_note": "STRONG written no-train; ONLY strong option with an India (Mumbai)
region for residency.",
    "fit_for_us": "medium"
  },
  {
    "name": "Lambda Labs (Lambda Cloud)",
    "what": "Pure ML-focused GPU cloud, simple on-demand pricing, free egress.",
    "pros": [
      "ML-native UX, pre-baked CUDA/PyTorch; A100 80GB ~$1.99-$2.49/hr, H100
~$3.29-$4.29/hr",
      "Free data egress (vs AWS $0.09/GB, GCP $0.12/GB)",
      "Good multi-GPU clusters for Option B"
    ],
    "cons": [
      "No spot/preemptible tier",
      "Privacy terms less explicit publicly - needs DPA review before the corpus goes on
it",
      "US-only, no India/Asia region; historically tight capacity"
    ],
    "cost_note": "A100 80GB ~$1.99-$2.49/hr, H100 ~$3.29-$4.29/hr, free egress. Our fine-
tune ~$5-$20.",
    "license": "Commercial cloud; you keep weights/data.",
    "privacy_note": "UNVERIFIED - obtain/read DPA before using for consented video.",
    "fit_for_us": "medium"
  },
  {
    "name": "Together AI (managed fine-tuning API)",
    "what": "Managed LoRA/full fine-tuning of open models priced per training-token; hosted
serving.",
    "pros": [
      "Lowest operational burden: managed LoRA ~$0.48/M tokens (<=16B); fastest path to a
first checkpoint",
      "Supports Qwen and 'any open-source model from HF Hub'; resume-from-checkpoint",
      "SOC2 Type II, HIPAA-aligned, Asia/ME deployment; 'data and models remain fully under
your ownership'"
    ],
    "cons": [
      "Could NOT confirm a verbatim no-train clause or public DPA - BLOCKED for the
consented corpus until signed DPA obtained",
      "Token pricing opaque for VIDEO/multimodal; Qwen2.5-VL video FT support is the open
question",
      "Managed = ship data to their cloud; weakest fit with 'data never leaves device'"
    ],
    "cost_note": "LoRA ~$0.48/M tokens (<=16B); ~$8-$12/M typical. Video token cost
unverified.",
    "license": "Commercial managed; you own resulting weights (per claim) - verify
downloadability.",
    "privacy_note": "UNCONFIRMED no-train clause + no public DPA -> BLOCKED for the corpus
until signed DPA + no-train assurance.",
    "fit_for_us": "weak"
  },
  {
    "name": "vast.ai (marketplace) - Secure tier only",
    "what": "Peer-to-peer GPU marketplace, cheapest rates. Community (unvetted) vs Secure
Cloud (ISO-27001+ vetted DCs).",
    "pros": [
      "Cheapest anywhere: A100 80GB ~$0.67/hr, H100 ~$1.87/hr, spot to ~$0.34/hr - great for
non-sensitive experiments",
      "Secure Cloud: vetted DCs (ISO 27001 min), SOC2 Type II platform-level, per-tenant
containers, instant data destruction"
    ],
    ],

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    "cons": [
        "Community/peer-to-peer tier unvetted - DIRECT conflict with our hard privacy requirement for consented video",
        "Price volatility; reliability varies host-to-host; no India region; rough operability"
    ],
    "cost_note": "A100 80GB ~$0.67/hr, H100 ~$1.87/hr (spot ~$0.34/hr). Cheapest by 2-5x - Secure tier only, ideally non-sensitive.",
    "license": "Marketplace; you keep weights/data.",
    "privacy_note": "Community tier FAILS the requirement. Secure Cloud acceptable for non-consented/synthetic data.",
    "fit_for_us": "weak"
},
{
    "name": "AWS SageMaker (EC2 P4d/P5) - Mumbai ap-south-1",
    "what": "Enterprise managed training or raw EC2 GPU. ap-south-1 (Mumbai) fully supported.",
    "pros": [
        "India residency: ap-south-1 full region with SageMaker + P4d (A100)",
        "VPC isolation by default, CMEK/KMS encryption, GDPR-aligned; can opt out of aggregated-metadata sharing",
        "P4/P5 price cuts up to 45% (2025); spot + savings plans; mature distributed training"
    ],
    "cons": [
        "High complexity + cost for a solo founder; SageMaker mgmt premium; spot interruptions",
        "Default aggregate library-usage metadata collection (opt-out is a footgun)",
        "Steep learning curve; heavy SDK lock-in; among the most expensive per-hour"
    ],
    "cost_note": "p4d (8xA100) ~$3.07-$3.43/hr/GPU-equiv; P5 (H100) ~$6.88/hr on-demand / ~$3.83 spot; plus SageMaker premium.",
    "license": "Commercial cloud; you keep weights/data.",
    "privacy_note": "STRONG (VPC+KMS+GDPR, India region) IF you opt out of metadata sharing and use VPC. Operationally heavy.",
    "fit_for_us": "weak"
},
{
    "name": "Azure ML - Central India",
    "what": "Enterprise managed ML on Azure NC/ND A100/H100 VMs; Central India region.",
    "pros": [
        "India residency: Central India with A100/H100",
        "Cheap spot: NC A100 v4 ~$0.82/hr spot vs ~$3.67/hr on-demand",
        "Enterprise compliance (GDPR/ISO/SOC), VPC/private endpoints, CMEK; good multi-GPU"
    ],
    "cons": [
        "Could NOT confirm an Azure ML 'we don't train on your data' statement - needs Microsoft DPA review",
        "On-demand H100 very expensive (~$6.98-$12/hr/GPU); only spot attractive",
        "Heavy enterprise tooling; high overhead; pipeline lock-in"
    ],
    "cost_note": "NC A100 v4 ~$3.67/hr on-demand / ~$0.82/hr spot; ND H100 v5 ~$6.98-$12/hr/GPU. India region available.",
    "license": "Commercial cloud; you keep weights/data.",
    "privacy_note": "Likely STRONG via Microsoft DPA + private endpoints, UNVERIFIED here - read the DPA first.",
    "fit_for_us": "weak"
},
{
    "name": "CoreWeave / Paperspace (DigitalOcean Gradient)",
    "what": "CoreWeave = GPU-specialist for large reserved fleets; Paperspace = simpler notebook/VM GPU cloud.",
    "pros": [
        "CoreWeave: SOC2/HIPAA/GDPR/ISO 27001; strong for very large multi-GPU H100 fleets (Option B if big)",
        "Paperspace: no minimum commitment, notebook-friendly (A100 ~$3.09/hr, H100 ~$5.95/hr

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promo)"
    ],
    "cons": [
        "CoreWeave enterprise/reserved-oriented (H100 ~$6.16/hr/GPU, A100 ~$2.70/hr) -
        overkill/pricey for small jobs",
        "Paperspace mid-to-high pricing, no spot; tight capacity",
        "Neither has an India region; neither beats Modal/RunPod on privacy specificity"
    ],
    "cost_note": "CoreWeave A100 ~$2.70/hr, H100 ~$6.16/hr/GPU. Paperspace A100 ~$3.09/hr,
    H100 ~$5.95/hr. No India region.",
    "license": "Commercial cloud; you keep weights/data.",
    "privacy_note": "CoreWeave has the cert stack; Paperspace less differentiated.
    Acceptable, not better than Modal/RunPod.",
    "fit_for_us": "weak"
  }
],
"key_sources": [
    "https://modal.com/docs/guide/security",
    "https://modal.com/pricing",
    "https://modal.com/blog/region-selection-launch",
    "https://www.runpod.io/pricing",
    "https://docs.runpod.io/references/security-and-compliance",
    "https://www.runpod.io/press/runpod-meets-hipaa-and-gdpr-standards",
    "https://cloud.google.com/blog/products/ai-machine-learning/google-cloud-generative-ai-
    data-residency-guarantees-for-data-stored-at-rest",
    "https://cloud.google.com/vertex-ai/pricing",
    "https://lambda.ai/pricing",
    "https://www.together.ai/fine-tuning",
    "https://docs.together.ai/docs/fine-tuning-pricing",
    "https://vast.ai/pricing",
    "https://docs.vast.ai/guides/reference/faq/security",
    "https://aws.amazon.com/sagemaker/pricing/",
    "https://docs.aws.amazon.com/sagemaker/latest/dg/data-protection.html",
    "https://azure.microsoft.com/en-us/pricing/details/machine-learning/",
    "https://learn.microsoft.com/en-us/azure/virtual-machines/sizes/gpu-
    accelerated/nca100v4-series",
    "https://www.coreweave.com/pricing",
    "https://www.spheron.network/blog/gpu-cloud-pricing-comparison-2026/",
    "https://willitrunai.com/can-run/qwen-2.5-7b-on-rtx-2070-super-8gb",
    "https://kaitechup.substack.com/p/qwen25-qlora-lora-and-full-fine-tuning"
],
"recommendation": "Decouple from raw GPU price: our fine-tune (Apache-2.0 Qwen2.5-VL-7B +
LoRA/QLoRA on 5k-20k clip-QA pairs) is a sub-24h, $10-$50 job on one A100 80GB, so privacy terms
+ operational ergonomics dominate. Plan: (1) DEV/CI on the RTX 2070 Super (8GB) - free; harness
plumbing, overfit-one-batch sanity, 4-bit 7B inference, NOT real training. (2) PRIMARY = Modal -
strongest written privacy contract ('Modal will never access your source code, inputs/outputs,
or stored data'; 7-day max retention; SOC2 Type II), per-second billing with scale-to-zero
matching the retrain-on-each-golden-batch cadence, Python/recipe-driven model that slots into a
one-command agent-runnable harness and the future ComputeBackend abstraction; run a thin glue
over ms-swift on Modal A100s. (3) FALLBACK / Option-B scale-up = RunPod Secure Cloud (A100
~$1.19-$1.29/hr, SOC2+GDPR+HIPAA, contractual no-inspect) - Secure Cloud only, never Community.
(4) KEEP Vertex AI / GCP on the bench: ONLY strong-privacy choice with an India (asia-south1
Mumbai) region + written no-train-on-customer-data - adopt if/when India residency or an audited
managed pipeline is mandated. (5) Use vast.ai/Lambda ONLY for non-sensitive synthetic/ablation
runs - never the consented corpus on vast.ai community tier. (6) HARD GATE before any consented
video reaches Together / Lambda / Azure: signed DPA + explicit written 'we do not train on or
retain your data' assurance; until then BLOCKED for the corpus. Sequencing: unblocks PLATFORM P5
design only - do not implement until P0-P4 (async jobs, storage/compute abstractions) ship, per
owner directive.",
    "confidence": "medium",
    "caveats": [
        "Pricing is volatile and sourced largely from aggregator blogs plus some primary pages;
        treat all dollar-per-hour figures as +/-15-25% and re-check live pricing pages before committing
        budget.",
        "Biggest UNVERIFIED gap: VIDEO/multimodal fine-tuning support and cost. Most data found is

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LLM-text-centric. Qwen2.5-VL VIDEO fine-tuning (frames as image tokens) can balloon token counts and VRAM far beyond text-7B numbers - the \$10-\$50/one-A100 estimate is for a modest LoRA run and must be re-benchmarked on real clip-QA data with actual frame counts before trusting it for Option B.",

"Together AI, Lambda Labs, and Azure ML privacy terms could NOT be confirmed to include an explicit no-train-on-your-data clause from public pages; flagged as needing a signed DPA before the corpus touches them.",

"Vertex/GCP nuance: the published gen-AI residency region list (excludes asia-south1) governs Google's MANAGED gen-AI APIs, not custom training on your own weights via Compute Engine GPUs - asia-south1 GPUs exist and the no-train principle applies, but confirm A100/H100 quota/capacity in asia-south1 (capacity-constrained).",

"RTX 2070 Super: 4-bit 7B text inference fits ~7.1GB and QLoRA training of a 7B can fit ~7GB in ideal text-only conditions, but a 7B-VLM with video frames realistically needs 16-24GB; treat the 2070S as dev/inference-only. Turing also lacks bf16/modern kernels.",

"India-region: only Vertex/GCP, AWS (ap-south-1), Azure (Central India) offer in-India residency among providers studied; Modal/RunPod/Lambda/Vast/CoreWeave/Paperspace do not. For batch TRAINING this is usually irrelevant, but if a legal residency-at-rest requirement emerges only the three hyperscalers qualify.",

"Compliance certs (SOC2/GDPR/HIPAA) attest to security controls, NOT a 'we won't train on your data' promise. The two clear written no-train/no-access assurances found are Modal (no-access) and Google/Vertex (no-train-without-permission); everything else needs contract review."

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]
},
{
  "dimension": "Model architecture for rally temporal localization/segmentation (action =
rally; labels = [start,end] intervals) over per-frame features",
  "summary": "For OUR situation ? per-frame features already in hand (WASB trajectory ~3-4
dims + optional audio/pose), only ~3-10 labeled matches, RTX 2070S/8GB, commercial intent, on-
device-capable ? the right model class is a LIGHT feature-input temporal model, NOT a heavy end-
to-end video network. Two viable families, both consuming pre-extracted per-frame feature
sequences (so no RGB pipeline required and they ride directly on WASB output):\n\n(1) TEMPORAL
ACTION LOCALIZATION (TAL) ? ActionFormer / TriDet / TemporalMaxer. These output (start, end,
class) instances directly via a multi-scale feature pyramid + classification/regression heads.
This is the most NATURAL fit for rally [start,end] + ending_reason: each rally is one detected
instance; ending_reason becomes the instance class (multi-class head) or a second head.
ActionFormer and TriDet are MIT and accept ARBITRARY input feature dimension via a learned input
projection (confirmed: the projection maps any D-dim feature to the model width), so the WASB
(X,Y,Visibility) trajectory can be fed directly ? I3D is NOT required, just the default choice
on benchmarks.\n\n(2) TEMPORAL ACTION SEGMENTATION (TAS) ? MS-TCN++ / ASFormer / DiffAct. These
do per-frame dense labeling (frame -> {rally, non-rally}); rally segments are recovered by
merging contiguous 'rally' frames. ASFormer is the standout for OUR data-hunger: it is
explicitly designed to train FROM SCRATCH on tiny datasets (GTEA = 28 videos, 11 classes) with
only ~1.1M params, by injecting local convolutional inductive bias to compensate for the
Transformer's missing inductive bias on small data. This is the closest published analogue to
our n=3-10 regime.\n\nKEY LICENSE FINDING (a real trap): the ORIGINAL ms-tcn repo (yabufarha/ms-
tcn) is MIT + Commons Clause = NON-commercial; do NOT vendor it. The MS-TCN++ repo (sj-li/MS-
TCN2), ASFormer (ChinaYi/ASFormer), DiffAct (Finspire13/DiffAct), ActionFormer
(happyharrycn/actionformer_release), and TriDet (dingfengshi/TriDet) are all clean MIT and
commercially usable. TemporalMaxer's LICENSE file could not be confirmed (404 on raw fetch) ?
treat as 'verify before vendoring'.\n\nBADMINTON PRECEDENT: published systems for the adjacent
task (per-frame hit detection from shuttle trajectory + pose) use exactly the light-sequence-
model recipe ? GRU-based recurrent nets over court/pose/shuttle features predicting per-frame
hit/no-hit (the MonoTrack lineage), and rule/threshold pipelines over TrackNet+YOLO. None use
heavy end-to-end video models for the localization step. This validates: a SMALL sequence model
over per-frame features is the field-standard approach for racket-sports temporal structure at
our data scale.\n\nNONE of these are the 'Option B' end-to-end multimodal VLM (Qwen-VL/Gemma)
from the grounding brief ? they are the bridge between today's heuristic+LogisticRegression and
that future VLM. They are the right FIRST owned model precisely because they learn from the
labels and features we ALREADY have, with no paid-LLM data and no large pretraining corpus.",
  "options": [
    {
      "name": "ASFormer (TAS) ? RECOMMENDED first owned model",
      "what": "Transformer for action segmentation that adds local convolutional inductive
bias (dilated temporal convs) to a vanilla Transformer so it trains from scratch on small data.
Per-frame dense classifier: every frame -> {rally, non-rally(, ending_reason)}; rally
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[start,end] recovered by merging contiguous rally frames + boundary cleanup. Input = per-frame
feature sequence of arbitrary dim (we feed WASB X,Y,Visibility +/- audio-onset +/- pose). ~1.1M
params.",
  "pros": [
    "Explicitly engineered for SMALL training sets ? trains from scratch on GTEA (28
videos); the only surveyed model whose paper makes the small-data claim its central thesis,
matching our n=3-10 regime",
    "Tiny (~1.1M params) -> trivially fits RTX 2070S 8GB; trains in minutes; on-
device/edge inference is easy",
    "Consumes per-frame features directly ? rides on existing WASB trajectory output; no
RGB/I3D extraction pipeline needed; trivially fuses audio-onset + pose by concatenating feature
channels (directly enables Option A modular fusion)",
    "Clean MIT (ChinaYi/ASFormer) ? commercially usable, fine-tunable, wrappable",
    "Per-frame formulation is forgiving of label noise and variable rally length;
ending_reason maps to extra per-frame classes or a per-segment head",
    "Beats our current ceiling plausibly: it LEARNS temporal structure rather than
threshold-tuning, and degrades gracefully where the LogisticRegression+windowing baseline
plateaued (~F1 0.73)"
  ],
  "cons": [
    "Per-frame segmentation needs a post-merge step to produce clean [start,end] intervals
(minor engineering; over/under-segmentation is the classic TAS failure mode)",
    "No native 'instance' concept ? two rallies separated by a 1-frame gap can merge;
needs a small gap/duration prior (we already have the net-crossing gate to reuse)",
    "Standard recipe assumes I3D features; using raw low-dim WASB features is off-the-
beaten-path and may need a feature-smoothing / windowing front-end to give the convs enough
signal",
    "Boundary precision is frame-granular, not sub-frame regressed (TAL models regress
boundaries more precisely)"
  ],
  "cost_note": "Free/OSS. Training cost ~minutes on the existing RTX 2070S; no cloud GPU
required at this data scale. Inference is CPU-feasible.",
  "license": "MIT (ChinaYi/ASFormer) ? verified, commercially clean",
  "privacy_note": "Fully local; ~1.1M-param model runs on-device, user video/features
never leave the box. No paid-LLM data anywhere in the loop.",
  "fit_for_us": "strong"
},
{
  "name": "ActionFormer (TAL) ? RECOMMENDED parallel/second model",
  "what": "Single-stage Transformer for temporal action localization. Multi-scale feature
pyramid + shared classification/regression heads emit (start, end, class) action instances
directly at every time step. Input = per-frame feature sequence; a learned input-projection
layer accepts ARBITRARY feature dimension (confirmed), so WASB trajectory feeds in without
I3D.",
  "pros": [
    "Most NATURAL label mapping: rally = one detected instance with regressed [start,end];
ending_reason = the instance class (multi-class head). Matches our CSV schema
(rally_number,start,end,ending_reason) almost 1:1",
    "Sub-frame boundary regression -> more precise [start,end] than frame-merged TAS,
directly attacking our F1 ceiling on boundary localization",
    "Input projection accepts any feature dim ? WASB (X,Y,Visibility) + audio + pose
concatenated; no RGB pipeline",
    "Clean MIT (happyharrycn/actionformer_release); strong, well-maintained, widely forked
codebase; TriDet/TemporalMaxer build on it so swapping later is cheap",
    "Standard, agent-runnable training recipe (config-driven) fits the 'one command +
report card' harness goal"
  ],
  "cons": [
    "More data-hungry than ASFormer ? designed/validated on THUMOS/ActivityNet (hundreds-
to-thousands of videos); benchmark papers do NOT claim few-shot, so at n=3-10 it risks
overfitting and may UNDERPERFORM ASFormer initially (same failure we saw when a learned model
lost to the baseline at n=2)",
    "Bigger than ASFormer (multi-scale pyramid + attention) though still small vs a VLM;
needs more careful regularization/augmentation on tiny data",
    "Default features are I3D; low-dim trajectory-only input is unproven and may need

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feature engineering to give attention enough to attend to",
    "NMS / instance de-duplication tuning adds knobs"
  ],
  "cost_note": "Free/OSS. Trains on RTX 2070S; somewhat heavier than ASFormer but still no
cloud GPU needed at our scale.",
  "license": "MIT (happyharrycn/actionformer_release) ? verified, commercially clean",
  "privacy_note": "Fully local, small model, on-device inference; no paid-LLM data.",
  "fit_for_us": "strong"
},
{
  "name": "TriDet (TAL)",
  "what": "One-stage TAL built ON TOP of ActionFormer's codebase. Adds a Trident-head
modeling the action boundary as a relative probability distribution + Scalable-Granularity
Perception (SGP) layer replacing some self-attention. Same (start,end,class) instance output and
same arbitrary-feature input as ActionFormer.",
  "pros": [
    "Better boundary modeling than ActionFormer (the Trident-head is purpose-built for
precise [start,end] ? directly relevant to our boundary-localization ceiling)",
    "Drop-in once ActionFormer is wired (shares ActionFormer's data/feature plumbing), so
it's a cheap upgrade path, not a separate integration",
    "Clean MIT (dingfengshi/TriDet); SGP reduces reliance on full self-attention -> a bit
lighter/faster",
    "Releases VideoMAEv2 feature configs ? useful if we later add an RGB cue"
  ],
  "cons": [
    "Same small-data risk as ActionFormer (benchmark-scale validation; no few-shot claim)
? not the right thing to START with at n=3-10",
    "More moving parts than ActionFormer; marginal gains may be invisible until the corpus
grows",
    "Off-benchmark low-dim trajectory input equally unproven here"
  ],
  "cost_note": "Free/OSS; same hardware envelope as ActionFormer.",
  "license": "MIT (dingfengshi/TriDet) ? verified",
  "privacy_note": "Fully local; no paid-LLM data.",
  "fit_for_us": "medium"
},
{
  "name": "MS-TCN++ (TAS)",
  "what": "Multi-stage temporal convolutional network: stage 1 makes an initial per-frame
prediction, refinement stages iteratively smooth it. Pure dilated temporal convs (no attention).
Per-frame {rally/non-rally} -> merge to intervals. ~0.8M params.",
  "pros": [
    "Smallest/simplest (~0.8M params, conv-only) -> fastest to train, easiest on 8GB,
strongest edge/on-device story",
    "Conv inductive bias handles small data reasonably (precursor to ASFormer's design);
multi-stage refinement directly fights over-segmentation",
    "Clean MIT on the CORRECT repo (sj-li/MS-TCN2)",
    "Battle-tested, minimal dependencies, easy to read/modify -> good fit for the
'minimal/readable harness' goal"
  ],
  "cons": [
    "LICENSE TRAP: the original yabufarha/ms-tcn repo is MIT + Commons Clause (NON-
commercial). MUST vendor sj-li/MS-TCN2 (clean MIT), not the original ? easy to get wrong",
    "Generally a notch below ASFormer in accuracy on the same benchmarks (no attention;
ASFormer was designed to beat it)",
    "Same per-frame->interval merge + over/under-segmentation concerns as all TAS",
    "Frame-granular boundaries, no sub-frame regression"
  ],
  "cost_note": "Free/OSS; cheapest to train/run of the whole set.",
  "license": "MIT ? but ONLY via sj-li/MS-TCN2. AVOID yabufarha/ms-tcn (Commons Clause,
non-commercial).",
  "privacy_note": "Fully local, tiny model, strong on-device fit; no paid-LLM data.",
  "fit_for_us": "medium"
},
{

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```

    "name": "Plain 1D-CNN / GRU / small Transformer over features (DIY sequence model)",
    "what": "A from-scratch per-frame or windowed sequence classifier (1D-CNN, BiGRU, or a
small Transformer) over WASB+audio+pose features, emitting per-frame rally prob ->
threshold+merge -> [start,end]. This is the direct learned successor to today's
windowing+LogisticRegression and matches the published badminton hit-detection recipe (BiGRU
over court/pose/shuttle).",
    "pros": [
        "Maximum control + minimum dependencies; trivially small, trivially on-device, trains
in seconds",
        "Matches the field-standard racket-sports recipe (MonoTrack-style GRU hit detector
over trajectory/pose) -> lowest-risk, known-to-work pattern at our data scale",
        "No license questions at all (our own code, our own weights)",
        "Easiest to make deterministic/reproducible for the agent-runnable harness; cleanest
baseline to A/B against ASFormer"
    ],
    "cons": [
        "We essentially already have this (distilled LogisticRegression) and it UNDERPERFORMED
the heuristic at n=2 ? a bare GRU may not clear the ~0.73 bar without the multi-stage refinement
/ inductive-bias tricks ASFormer/MS-TCN++ bake in",
        "Reinventing temporal-modeling tricks (multi-scale, refinement, boundary regression)
that ASFormer/ActionFormer already implement and validate",
        "No community-tuned recipe -> more of our own tuning time; easy to under-build"
    ],
    "cost_note": "Free; negligible compute. Mostly costs our engineering time.",
    "license": "N/A (own code) ? no third-party weights/data",
    "privacy_note": "Fully local; no paid-LLM data.",
    "fit_for_us": "medium"
},
{
    "name": "TemporalMaxer (TAL)",
    "what": "Parameter-free max-pooling-only TAL on the ActionFormer codebase: drops self-
attention for local max-pooling. Outputs (start,end,class) instances. ~2.8x fewer GMACs and ~3x
faster inference than ActionFormer.",
    "pros": [
        "Cheapest/fastest TAL at inference (best edge/on-device latency of the TAL family);
fewer params than ActionFormer",
        "Shares ActionFormer's data/feature plumbing -> cheap to try alongside it",
        "Claims to beat ActionFormer on several TAL benchmarks despite simplicity"
    ],
    "cons": [
        "LICENSE UNCONFIRMED ? could not fetch a LICENSE file (raw 404); must verify it is
MIT/Apache before any commercial use",
        "Same benchmark-scale validation as ActionFormer; no few-shot claim -> not a starting
choice at n=3-10",
        "Max-pooling-only may lose the fine temporal context that matters for rally-boundary
precision on noisy amateur footage"
    ],
    "cost_note": "Free/OSS; lowest inference cost of the TAL set.",
    "license": "UNVERIFIED (likely permissive given ActionFormer base, but confirm the
LICENSE file before relying on it)",
    "privacy_note": "Fully local; no paid-LLM data.",
    "fit_for_us": "weak"
},
{
    "name": "DiffAct (TAS, diffusion)",
    "what": "Diffusion-based action segmentation: iteratively denoises per-frame action
predictions conditioned on video features. ICCV 2023. Per-frame {rally/non-rally} -> merge.",
    "pros": [
        "SOTA-class accuracy on standard TAS benchmarks; principled boundary refinement via
denoising",
        "Clean MIT (Finspire13/DiffAct)",
        "Same per-frame feature input as MS-TCN++/ASFormer -> would consume WASB features"
    ],
    "cons": [
        "Most complex + most data-hungry of the TAS set; diffusion training is finicky and the

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LAST thing you want at n=3-10",
    "Iterative denoising = slower inference -> weakest on-device story among TAS",
    "No few-shot / small-data claim; overkill until the golden corpus is large",
    "Highest harness/operability burden vs the 'minimal/readable, deterministic' goal"
],
"cost_note": "Free/OSS but heaviest to train and tune; multi-step inference.",
"license": "MIT (Finspire13/DiffAct) ? verified",
"privacy_note": "Fully local; no paid-LLM data (but heavier model).",
"fit_for_us": "weak"
}
],
"key_sources": [
    "https://arxiv.org/abs/2202.07925",
    "https://github.com/happyharrycn/actionformer_release",
    "https://github.com/dingfengshi/TriDet",
    "https://github.com/dingfengshi/TriDet/blob/master/LICENSE",
    "https://arxiv.org/abs/2303.09055",
    "https://github.com/TuanTNG/TemporalMaxer",
    "https://github.com/sj-li/MS-TCN2",
    "https://github.com/sj-li/MS-TCN2/blob/master/LICENSE",
    "https://github.com/yabufarha/ms-tcn/blob/master/LICENSE",
    "https://github.com/ChinaYi/ASFormer",
    "https://ar5iv.labs.arxiv.org/html/2110.08568",
    "https://arxiv.org/abs/2110.08568",
    "https://github.com/Finspire13/DiffAct",
    "https://github.com/Finspire13/DiffAct/blob/main/LICENSE",
    "https://arxiv.org/pdf/2006.09220",
    "https://pmc.ncbi.nlm.nih.gov/articles/PMC11244353/",
    "https://ar5iv.labs.arxiv.org/html/2204.01899"
],
"recommendation": "FIRST owned model = ASFormer (MIT, ChinaYi/ASFormer), as a TEMPORAL ACTION SEGMENTATION model over our existing per-frame features. Rationale: it is the ONLY surveyed architecture whose central published result is training FROM SCRATCH on a tiny dataset (GTEA, 28 videos) with ~1.1M params, by adding convolutional inductive bias to compensate for small-data overfitting ? exactly the failure mode we hit when a learned model lost to the heuristic at n=2. It consumes per-frame feature sequences directly, so it rides on the WASB trajectory we already produce (X,Y,Visibility) and trivially fuses audio-onset and pose by concatenating channels ? which simultaneously delivers the Option A modular-fusion milestone. It is tiny (fits the RTX 2070S, trains in minutes, runs on-device), clean MIT, fully local (no paid-LLM data, preserving the privacy lever and all guardrails), and learns temporal structure rather than threshold-tuning, giving it a credible path past the ~F1 0.73 ceiling.\n\nConcrete plan:\n1. Map labels: per-frame target = rally(1)/non-rally(0) from the rally [start,end] CSVs; recover predicted intervals by merging contiguous rally frames + a min-gap/min-duration prior (reuse the existing net-crossing gate). Treat ending_reason as a SECOND, segment-level head (5 classes) once interval detection works ? do NOT try to learn it jointly first.\n2. Features: feed WASB (X,Y,Visibility) per frame; add audio-onset and pose channels as they come online (this IS Option A fusion). Smooth/window features so the dilated convs have signal on amateur 60fps footage.\n3. Split DISCIPLINE: split train/eval by MATCH, never by clip (prevents the leakage the brief flags). Gemini silver labels stay eval/teacher-only, never in training.\n4. Gate: ship only if held-out per-match F1 beats the heuristic+LogisticRegression baseline (~0.73) on the golden set ? the nanochat-style pass/fail verdict.\n\nPARALLEL second model = ActionFormer (MIT, happyharrycn/actionformer_release) as the TAL formulation: rally = one regressed (start,end,class=ending_reason) instance ? the cleanest 1:1 map to our CSV schema and the better long-term boundary-precision play. Run it as the A/B challenger; expect it to LOSE to ASFormer at n=3-10 (it has no few-shot claim and is benchmark-scale) but to OVERTAKE once the Roora-fed golden corpus grows into the hundreds of matches. TriDet is the cheap drop-in boundary-precision upgrade on top of ActionFormer for that later stage.\n\nEXPLICITLY DEFER: DiffAct and TemporalMaxer (too data-hungry / license-unconfirmed for a first model). And note these models are the BRIDGE to the brief's Option B end-to-end Qwen-VL/Gemma VLM ? not a replacement for it; they unblock P5 design using only data and features we already own.\n\nLICENSE GUARDRAIL TO RECORD IN COMMERCIALIZATION.md: if MS-TCN++ is ever chosen, vendor ONLY sj-li/MS-TCN2 (clean MIT) ? the original yabufarha/ms-tcn carries a Commons Clause (non-commercial) and MUST NOT be used.",
"confidence": "high",
"caveats": [

```

"TemporalMaxer's license is UNCONFIRMED ? the raw LICENSE fetch 404'd and search did not surface a license. It is built on ActionFormer (MIT) so it is likely permissive, but verify the actual LICENSE file before any commercial vendoring. I marked it 'weak' partly for this reason.",

"LICENSE TRAP confirmed by hand: yabufarha/ms-tcn = MIT + Commons Clause (NON-commercial); sj-li/MS-TCN2 (the MS-TCN++ repo) = clean MIT. These are DIFFERENT repos for what is colloquially called 'MS-TCN' ? vendor the right one.",

"The few-shot evidence for ASFormer is GTEA = 28 videos, which are MULTI-CLASS densely-labeled cooking videos, not 3-10 single-action sports matches. The small-data claim is well-supported but our regime is even smaller and differently structured; expect to need feature smoothing, augmentation (time-warp/jitter on trajectories), and possibly leave-one-match-out CV. A from-scratch learned model still may not beat the ~0.73 heuristic until the golden corpus grows ? the brief itself notes the bottleneck is golden DATA + WASB recall, not the method. This recommendation lowers the floor and unblocks design; it does not guarantee a win at n=3-10.",

"All TAL/TAS benchmark recipes default to I3D (2048-d) features. Feeding raw ~3-4-dim WASB trajectory is OFF the validated path. ActionFormer's input-projection and ASFormer's input layer both accept arbitrary feature dim (so it is mechanically supported), but accuracy on such low-dim input is unproven in the literature ? a learned feature front-end or hand-built feature stack (velocity, acceleration, visibility-run-length, net-crossing flags) will likely be needed.",

"ending_reason (winner/forced_error/unforced_error/service_fault/let) is a SEMANTIC per-rally label that trajectory features alone almost certainly cannot predict well; treat it as a separate, later head and do not let it gate the core [start,end] detection milestone.",

"I did not benchmark these models against each other on OUR data (cannot run GPU/torch here per the environment note). The ranking is from architecture properties, published small-data behavior, and license ? empirical A/B on the golden set is still required and is the actual ship/no-ship gate.",

"These models are NOT the brief's Option B end-to-end multimodal VLM (Qwen-VL/Gemma). They are the intermediate owned model that beats the heuristic baseline using current assets; the VLM remains the 12-24mo target after scaffolding + corpus health, per the roadmap."

```
]
},
{
  "dimension": "OPTION B ? Owned end-to-end multimodal model: open, video-capable
Apache-2.0/MIT bases fine-tunable for rally temporal grounding (frames+audio+pose ? rally
[start,end]).",
```

```
  "summary": "For an OWNED, on-device-capable rally-temporal-segmentation model we must split
candidates into two classes. (1) BASES we can legally own and ship commercially ? the only
viable path ? are generic video-VLMs with clean Apache-2.0/MIT licenses, NOT the temporal-
grounding-specialized research models. (2) The temporal-grounding specialists named in the brief
(TimeChat, VTimeLLM, Momensor) and the classic video encoders (VideoMAE) are LICENSE-POISONED
for a product: they are research-only / CC-BY-NC / LLaMA-derived-with-no-LICENSE, so they are
usable ONLY as METHOD references (how to tokenize timestamps, how to format moment-localization
QA), never as the weights we fine-tune and serve. Top approved bases: Qwen2.5-VL-7B (Apache-2.0,
native timestamp grounding via mRoPE+absolute-time-alignment ? best-documented temporal-
grounding VLM in the legal set), Qwen3-VL-4B/8B (Apache-2.0, explicit Text?Timestamp Alignment,
4B fits the on-device/privacy lever), Gemma 4 4B/12B (Apache-2.0 ? VERIFIED, a genuine reversal
from Gemma 1/2/3's restrictive terms, native frames+audio+video without separate encoders, ideal
for the Option-B multimodal fusion), and InternVL3-8B (MIT, built on Qwen2.5-7B Apache base).
CRITICAL GUARDRAIL CONFIRMATIONS: Gemma 3 stays REJECTED (Gemma Terms of Use, viral) but Gemma 4
is Apache-2.0 and clean; Qwen2.5-VL-3B (Qwen Research, non-commercial) and Qwen2.5-VL-72B
(100M-MAU cap) are REJECTED per-size ? only the 7B and 32B Qwen2.5-VL sizes are Apache-2.0.
COMPUTE REALITY for our situation: the lone 8GB RTX 2070 Super CANNOT train any 7B-class video
VLM (video sequences push even QLoRA to ~20-24GB); it is an inference/eval box only. Training
MUST be a rented GPU (one 24-48GB A100/L40S-hour-class job). DATA REALITY: 128 human-labeled
rallies (96+32) is FAR below the 5k-20k clip?QA band needed for end-to-end fine-tuning ? so
Option B is correctly a 12-24mo horizon; the near-term move is LoRA on a 4B Apache base as a PoC
once Roora/rally-annotator grow the corpus, with the modular Option-A fusion shipping first.",
```

```
  "options": [
    {
      "name": "Qwen2.5-VL-7B-Instruct",
      "what": "7B generic video-VLM; native video temporal grounding (mRoPE + absolute-time
alignment lets it pinpoint relevant video segments / emit timestamps). The single best-
documented temporal-grounding-capable model inside the Apache-2.0 fence. Built on the Qwen2.5-7B
LLM with an optimized ViT (SwiGLU+RMSNorm).",
```

```

    "pros": [
        "EXACT license = Apache-2.0 at the 7B size (verified on the HF model card) ?
commercial-clean, derivative-wrappable, plugs into the provider registry with no MAU cap",
        "Native timestamp/moment-localization ability out of the box ? directly the rally
[start,end] task class; minimizes the grounding behavior we must teach",
        "Massive ms-swift / LoRA / QLoRA tooling + community fine-tuning recipes (the brief's
chosen engine targets exactly this family)",
        "Mature, stable checkpoint (not a research preview) ? safe to standardize the harness
on"
    ],
    "cons": [
        "7B cannot train on our 8GB RTX 2070 Super; LoRA needs ~20GB and even QLoRA on VIDEO
sequences lands ~24GB ? training is rented-GPU only",
        "7B is heavy for the on-device privacy lever; local INFERENCE at 1080p/60fps video is
feasible but not snappy on 8GB (expect quantized + low-fps sampling)",
        "Newer Qwen3-VL supersedes it on temporal alignment; choosing 7B trades cutting-edge
accuracy for a battle-tested baseline",
        "Still needs 5k-20k clip?QA pairs to beat the F1?0.73 heuristic ceiling ? base license
being clean does not solve the data scarcity"
    ],
    "cost_note": "Inference: runs locally on the 2070 Super quantized (8GB) for eval;
comfortable on a 16GB+ rented card. Training: one LoRA run ~4-12h on a 24GB rented GPU
(RunPod/Lambda L40S/A100 class), low-tens-of-USD per run.",
    "license": "Apache-2.0 (verified on huggingface.co/Qwen/Qwen2.5-VL-7B-Instruct). NOTE:
3B=Qwen Research (non-commercial, REJECT), 72B=Qwen License (100M-MAU cap, REJECT). Only 7B and
32B are Apache-2.0.",
    "privacy_note": "Local inference satisfies the privacy lever (video never leaves device)
but 8GB is the binding constraint; per-user LoRA adapters are tiny and deletable (drop the
adapter = delete the user's training).",
    "fit_for_us": "strong"
},
{
    "name": "Qwen3-VL-4B-Instruct (and 8B)",
    "what": "Newer Qwen VLM (released 2025-10) with explicit 'Text?Timestamp Alignment'
(evolves beyond T-RoPE) for precise timestamp-grounded event localization; second-level video
indexing, 256K-1M context. 4B is the small, on-device-friendly variant.",
    "pros": [
        "EXACT license = Apache-2.0 at BOTH 4B and 8B (verified on HF) ? commercial-clean",
        "4B is the best fit for the on-device/privacy lever: small enough to LoRA on a modest
rented 24GB card and to run quantized locally for inference",
        "Temporal grounding is a headline feature (explicit textual-timestamp alignment) ?
purpose-built for rally [start,end] emission",
        "Newest of the approved family ? strongest temporal modeling per the technical report;
future-proofs the base lock"
    ],
    "cons": [
        "Still cannot train on the 8GB local box (4B QLoRA on video is ~12-16GB+); training is
rented-GPU only, though cheaper than 7B",
        "Younger checkpoint ? fewer hardened community fine-tuning recipes than Qwen2.5-VL-7B
(more harness debugging on us)",
        "'Thinking' variants add latency; the Instruct variant is the right pick for
deterministic [start,end] output",
        "Same data-scarcity ceiling: 128 rallies is far below the training band"
    ],
    "cost_note": "Cheapest train of the strong options: 4B LoRA fits a single 24GB rented
GPU comfortably, likely <$10/run. Local quantized inference plausible on the 2070 Super.",
    "license": "Apache-2.0 (verified: Qwen3-VL-4B-Instruct and 8B-Instruct on HF).
Larger/MoE Qwen3-VL sizes must be re-checked per-size before any lock.",
    "privacy_note": "Best on-device candidate for the privacy lever given the 4B size; per-
user adapters deletable.",
    "fit_for_us": "strong"
},
{
    "name": "Gemma 4 (4B / 12B, multimodal)",
    "what": "Google's 2026 open multimodal family (2B/4B/12B/26B/31B). Processes

```

```

text+image+audio+VIDEO natively without separate encoder networks; 12B runs on a 16GB laptop.
The native frames+audio path maps directly onto our Option-B fusion goal (frames+audio+pose ?
rally).",
  "pros": [
    "EXACT license = Apache-2.0 ? VERIFIED, and a genuine reversal: Gemma 1/2/3 were under
the restrictive 'Gemma Terms of Use', Gemma 4 (from the 2026-04 launch) is true Apache-2.0 with
full commercial rights. This directly clears the guardrail that REJECTS Gemma 3",
    "Native audio+video in ONE model is uniquely aligned with Option B (collapse WASB-
trajectory + audio-hit + pose into a single owned model) ? fewer modality bridges to build",
    "4B variant is genuinely on-device (16GB laptop / quantized smaller) ? strong privacy
lever",
    "Backed by Google's tooling + ecosystem; broad fine-tuning support expected"
  ],
  "cons": [
    "Temporal-grounding/timestamp-emission is NOT as explicitly documented as Qwen's ?
we'd likely have to teach moment-localization output format more from scratch",
    "Very new (mid-2026) ? fine-tuning recipes for the temporal-grounding task are
immature; more harness risk",
    "Must KEEP verifying per-checkpoint: the brief's rejection was specifically Gemma 3;
do not let 'Gemma' shorthand leak a Gemma-3 weight into the pool",
    "Audio-native is a future-facing bet; our near-term audio cue (E1) is AMBER, so the
multimodal advantage is partly latent"
  ],
  "cost_note": "4B trains cheaply on a rented 24GB card; 12B runs locally for inference on
a 16GB machine (our 8GB box would need the 2B/4B quantized).",
  "license": "Apache-2.0 (VERIFIED across multiple 2026 sources; replaces Gemma Terms of
Use used by Gemma 1/2/3). Per-checkpoint re-verify before lock ? only Gemma 4 is clean; Gemma 3
remains REJECTED.",
  "privacy_note": "4B on-device satisfies the privacy lever; native local audio+video
keeps the whole multimodal pipeline on-device ? strongest privacy story of all options.",
  "fit_for_us": "strong"
},
{
  "name": "InternVL3-8B",
  "what": "OpenGVLab VLM, MIT-licensed wrapper built on a Qwen2.5-7B Apache-2.0 LLM +
InternViT vision encoder; supports interleaved image/video/text and extended video training
samples.",
  "pros": [
    "EXACT license = MIT on the InternVL3 wrapper, and the underlying Qwen2.5-7B base is
Apache-2.0 ? dual clean, commercial-safe",
    "Strong general video understanding; competitive VLM benchmarks",
    "MIT is the most permissive option ? zero ambiguity for derivative-wrapping",
    "Same Qwen2.5 lineage means ms-swift recipes largely transfer"
  ],
  "cons": [
    "Temporal-grounding/timestamp localization is LESS first-class than
Qwen2.5-VL/Qwen3-VL (general video QA, not purpose-built moment localization) ? more to teach
for rally [start,end]",
    "Per-SIZE base differs: 9B uses InternLM3 and 38B/78B use Qwen2.5-32B/72B ? the
72B-based 78B inherits the 100M-MAU cap; ONLY the 1B/2B/8B/14B/38B (Qwen2.5 ?32B) sizes are
clean. Verify per size before lock",
    "8B has the same 8GB-box training impossibility as Qwen2.5-VL-7B",
    "Adds a vision-encoder dependency (InternViT) to the stack"
  ],
  "cost_note": "Comparable to Qwen2.5-VL-7B: rented-GPU LoRA, local quantized inference
only.",
  "license": "MIT (InternVL3 wrapper) + Apache-2.0 (Qwen2.5 base) for the ?8B/14B/38B
sizes. REJECT InternVL3-78B (Qwen2.5-72B base = 100M-MAU cap). Verify per size.",
  "privacy_note": "Local inference feasible quantized; same adapter-deletion model as
Qwen.",
  "fit_for_us": "medium"
},
{
  "name": "VideoLLaMA2 / VideoLLaMA3 (DAMO)",
  "what": "DAMO video-LLMs; v3 (Jan 2025) is built on Qwen2.5 (7B and 1.5B/2B), code

```

Apache-2.0, with demonstrated temporal-grounding capability. v2 added explicit audio understanding (relevant to Option B fusion).",

```
"pros": [
  "CODE is Apache-2.0; v3 base LLM (Qwen2.5) is Apache-2.0; demonstrated temporal-
grounding notebook ? closest 'specialist behavior on a clean base' in the set",
  "v2's audio-understanding branch is a useful reference for our frames+audio fusion
target",
  "2B (Qwen2.5-1.5B) variant is small enough for the on-device lever"
],
"cons": [
  "BLOCKER: the released VideoLLaMA3 SERVICE/checkpoint is declared a 'research preview
intended for non-commercial use ONLY', subject to extra Qwen + data-source terms ? so we CANNOT
ship DAMO's fine-tuned weights. We could only re-train from the Apache Qwen2.5 base ourselves
using their (Apache) code, i.e. it collapses to 'use Qwen2.5-VL/Qwen3-VL + their training
recipe'",
  "Therefore not a true 'owned base' ? it's a method/recipe reference plus an Apache
code dependency",
  "Mixed signal: 'Apache code' vs 'non-commercial weights' is exactly the trap to avoid
asserting as clean"
],
"cost_note": "Same rented-GPU training profile; no advantage over directly using
Qwen3-VL/Gemma 4 as the base.",
"license": "Code Apache-2.0; RELEASED WEIGHTS = research/non-commercial preview (NOT
shippable). Use as recipe reference only.",
"privacy_note": "Moot for product use given the non-commercial weights; if re-trained
from Qwen2.5, privacy story = same as Qwen options.",
"fit_for_us": "weak"
},
{
  "name": "TimeChat / VTimeLLM / Momentor (temporal-grounding specialists)",
  "what": "Academic video-LLMs purpose-built for temporal grounding / moment localization.
TimeChat (CVPR'24, time-sensitive), VTimeLLM (CVPR'24, 'grasp video moments'), Momentor (fine-
grained temporal reasoning, trained on Moment-10M).",
  "pros": [
    "State-of-the-art METHODS for exactly our task: timestamp tokenization, temporal
position encoding (Momentor's TPM), moment-localization instruction formats ? high-value as
design references for our QA-pair schema and the grounding head",
    "Papers/datasets (e.g. Moment-10M format) inform how to convert rally CSVs ? clip?QA
pairs"
  ],
  "cons": [
    "LICENSE-POISONED as bases: TimeChat = research-preview non-commercial; VTimeLLM = CC-
BY-NC-ND 4.0 (NonCommercial + NoDerivs ? cannot even fine-tune/redistribute); Momentor = built
on LLaMA with NO LICENSE file in-repo (legally unusable + LLaMA's own restrictions). NONE are
Apache-2.0/MIT",
    "Cannot be the owned, commercial, fine-tunable base under our hard guardrail ? full
stop",
    "Older LLM backbones (Vicuna/LLaMA-class) lag the approved Qwen3-VL/Gemma 4 bases on
raw capability anyway"
  ],
  "cost_note": "N/A as a base (cannot ship). Zero-cost as reading/method references.",
  "license": "TimeChat = research-only/non-commercial; VTimeLLM = CC-BY-NC-ND 4.0;
Momentor = no LICENSE + LLaMA-derived. ALL REJECTED as bases; method-reference only.",
  "privacy_note": "N/A ? not deployable.",
  "fit_for_us": "weak"
},
{
  "name": "Video encoders: InternVideo2 / VideoMAE",
  "what": "Pure spatiotemporal video encoders (not generative VLMs) usable as a backbone
for a custom temporal-localization head (e.g. encoder ? CG-DETR-style grounding head, the
Option-A/modular or a from-scratch e2e route that avoids a full VLM).",
  "pros": [
    "InternVideo2-Stage2_6B = MIT (VERIFIED) and InternVideo2.5_Chat_8B = Apache-2.0 ?
clean encoder weights we CAN use commercially",
    "An encoder+grounding-head is far CHEAPER to train than a full VLM and could run on
```

the 8GB box at smaller sizes ? a realistic near-term owned-model PoC at our data scale",

"InternVideo2 is a proven temporal-grounding backbone (evaluated on QVHighlights/Charades-STA via CG-DETR)",

"Sidesteps the 'generative VLM is huge' problem for the privacy/on-device lever"

],

"cons": [

"VideoMAE weights = CC-BY-NC 4.0 ? REJECTED for commercial use (research-only); only its Apache-2.0 code components are reusable, not the pretrained checkpoints",

"Encoder-only path is NOT the generative video-QA Option-B end-state ? it answers 'where is the rally' but not the rich rally semantics (ending_reason etc.); it's a complementary/bridge architecture, not the VLM the brief ultimately wants",

"Building+maintaining a custom grounding head is extra engineering vs. fine-tuning an off-the-shelf VLM",

"Need to confirm InternVideo2's largest checkpoints per-file (some sub-weights vary)"

],

"cost_note": "Cheapest training of any option; a small InternVideo2 encoder + light head is plausibly trainable on the 8GB box or a single small rented card ? the only option that fits TODAY's 128-rally data scale.",

"license": "InternVideo2-Stage2_6B = MIT; InternVideo2.5_Chat_8B = Apache-2.0 (both VERIFIED, usable). VideoMAE checkpoints = CC-BY-NC 4.0 (REJECTED); VideoMAE code Apache-2.0 only.",

"privacy_note": "Small encoders are the most on-device-friendly option ? strongest fit for the local privacy lever at small data scale.",

"fit_for_us": "medium"

}

],

"key_sources": [

"https://huggingface.co/Qwen/Qwen2.5-VL-7B-Instruct",

"https://huggingface.co/Qwen/Qwen2.5-VL-72B-Instruct/blob/main/LICENSE",

"https://huggingface.co/Qwen/Qwen2.5-VL-3B-Instruct/blob/main/LICENSE",

"https://huggingface.co/Qwen/Qwen3-VL-8B-Instruct",

"https://huggingface.co/Qwen/Qwen3-VL-4B-Instruct",

"https://github.com/qwenlm/qwen3-vl",

"https://blog.google/innovation-and-ai/technology/developers-tools/gemma-4/",

"https://www.infoq.com/news/2026/04/google-gemma4/",

"https://venturebeat.com/technology/googles-new-open-source-gemma-4-12b-analyzes-audio-video-and-runs-entirely-locally-on-a-typical-16gb-enterprise-laptop",

"https://huggingface.co/OpenGVLab/InternVL3-8B",

"https://huggingface.co/OpenGVLab/InternVideo2-Stage2_6B",

"https://huggingface.co/OpenGVLab/InternVideo2_5_Chat_8B",

"https://github.com/DAMO-NLP-SG/VideoLLaMA3",

"https://github.com/DAMO-NLP-SG/VideoLLaMA2/blob/main/LICENSE",

"https://github.com/RenShuhuai-Andy/TimeChat",

"https://github.com/huangb23/VTimeLLM",

"https://github.com/DCDmlm/Momentor",

"https://huggingface.co/MCG-NJU/videomae-base",

"https://arxiv.org/html/2403.15377v1",

"https://arxiv.org/html/2402.11435v1",

"https://www.labellerr.com/blog/fine-tune-qwen-2-5-vl/",

"https://kaitchup.substack.com/p/qwen3-vl-fine-tuning-on-your-computer"

],

"recommendation": "LOCK the base as a two-stage decision, not one model. (1) NEAR-TERM PoC base = Qwen3-VL-4B-Instruct (Apache-2.0, explicit Text?Timestamp Alignment, smallest strong on-device candidate) as the primary, with Qwen2.5-VL-7B-Instruct (Apache-2.0, best-documented temporal grounding, most mature ms-swift recipes) as the de-risking fallback. Both are clean for commercial ownership and the privacy lever. (2) WATCH/EVALUATE Gemma 4 4B (Apache-2.0, VERIFIED reversal from the rejected Gemma 3) as the Option-B end-state base ? its native frames+audio+video in one model is the best structural match for collapsing WASB-trajectory + audio + pose into a single owned model; lock it for Option B once a fine-tuning recipe for timestamp emission is proven. DO NOT use TimeChat/VTimeLLM/Momentor/VideoMAE as bases (all non-commercial or no-LICENSE) ? mine them only for the timestamp-tokenization and clip?QA-format methods. DO NOT touch Qwen2.5-VL-3B (Qwen Research), Qwen2.5-VL-72B / InternVL3-78B (100M-MAU cap), DAMO's released VideoLLaMA3 weights (research-preview), or any Gemma ?3. COMPUTE: treat the 8GB RTX 2070 Super as an inference/eval-only box; budget a single rented 24-48GB GPU (RunPod/Lambda L40S or A100 class) for every fine-tune ? even QLoRA on video exceeds 8GB.

SEQUENCING (matches the brief's directive): ship modular Option-A fusion first; do NOT start any owned-model fine-tune until (a) the async job model / ComputeBackend P0 lands and (b) the golden corpus crosses ~thousands of clip?QA pairs (Roora + rally-annotator). At today's 128 rallies the only owned-model experiment that is data-realistic NOW is a small InternVideo2-encoder (MIT) + light grounding head as a bridge PoC ? the full VLM fine-tune is correctly a 12-24mo Option-B horizon.",

"confidence": "high",

"caveats": [

"Licenses change per-checkpoint and per-version. Re-verify the EXACT license file on the specific HF revision before any code lands ? especially: Qwen2.5-VL sizes (only 7B/32B Apache; 3B/72B rejected), InternVL3 sizes (78B inherits Qwen2.5-72B MAU cap), and ANY Gemma weight (only Gemma 4 is Apache; Gemma 3 stays REJECTED). The 'Gemma' shorthand is a live foot-gun.",

"Gemma 4's Apache-2.0 status is confirmed across multiple 2026 secondary sources and the official Google blog, but I did not open the raw LICENSE file on each Gemma 4 HF checkpoint ? do that before locking it as the Option-B base.",

"Compute/VRAM numbers are from community fine-tuning write-ups (Labellerr, Kaitechup, Unsloth), not a controlled benchmark on OUR footage. Real 1080p/4K-60fps video sequences with long context can push VRAM higher than the text/OCR LoRA figures cited; treat 24GB as a floor for a 7B video LoRA and validate empirically.",

"The 5k-20k clip?QA training band is the brief's own estimate; the true minimum to beat the F1?0.73 heuristic with an end-to-end VLM is unproven for this niche task and could be higher (or, with strong base priors + good QA formatting, somewhat lower). The honest n=2 result (learned model underperforms baseline) is the binding evidence that data ? not base choice ? is the current bottleneck.",

"Momentor having no LICENSE file is from a single fetch of the repo root; a LICENSE may exist in a subdir or have been added since ? but absence-of-license + LLaMA derivation is enough to disqualify it as a base regardless.",

"I treated 'on-device' against the stated 8GB RTX 2070 Super. If the owner provisions a 16-24GB local card later, the local-training calculus changes materially (4B QLoRA becomes locally trainable)."

]

},

{

"dimension": "LABEL & DATA REUSE / DATA ENGINEERING ? maximizing the value of our human rally CSVs + ending_reason + WASB trajectories across model generations (heuristics ? TAL ? owned video-VLM \"Option B\")",

"summary": "Our labels are far more valuable than their raw count (96 + 32 rallies ? 2 matches) suggests, IF we stop treating them as one-format eval files and start treating them as a model-agnostic \"labeled-corpus\" that we transcode into every consumer. (a) FORMATS: the rally [start,end]+ending_reason+sport rows already ARE the ActivityNet/ActionFormer schema modulo a transcode ? a `segment:[start,end]` (seconds) + `label` + per-video `duration/fps/subset` JSON; the SAME rows transcode into (i) candidate-window labels for our heuristic/LogReg distillation (already done via `training.label\_candidates`), (ii) ActivityNet-style JSON for any TAL net (ActionFormer/TriDet), and (iii) clip?QA text pairs (\"When does each rally happen?\" ? \"[start,end]?\") for the Option-B video-VLM. ending_reason becomes a second QA task for free. WASB trajectories are an auxiliary feature stream keyed by the same (video_id, frame). (b) AMPLIFICATION ? five legitimate levers that never touch paid-LLM outputs: (1) teacher-student / pseudo-labeling where the TEACHER is OUR OWN model or an open MIT model (WASB+heuristics, or a TAL net trained on the few labels), self-training on our ~26 GB unlabeled corpus ? SS-TAL is an active field and PLR/ALQA-style confidence-gated pseudo-labels are the standard recipe; (2) temporal data augmentation (speed/time-warp $\pm 20\%30\%$, temporal jitter, reverse, segment cut-mix/mixup) that multiplies effective segment count for any TAL/VLM trainer while preserving boundaries; (3) active learning ? spend the Roora budget where the model is least certain (boundary entropy / disagreement vs WASB), measurably beating random labeling under a fixed budget; (4) timestamp/point supervision for cheaper future labels (~? the time of full segments, SkipTag ~?, reaching near-full-supervision accuracy) ? relevant for ending_reason and for scaling new sports; (5) for Option B specifically, RL/RFT (GRPO) is dramatically more LABEL-efficient than SFT for temporal grounding (VideoChat-R1: +31.8 temporal-grounding with \"limited samples\", and quality-filtered 13K beat unfiltered 56K in VTG-R1) ? meaning our small hand-verified set is the right asset class. (c) HOW MANY MATCHES: heuristic threshold tuning needs ~3?5 matches for stable cross-val (we're under that at n?2, which is exactly why the learned model underperforms baseline); a from-scratch TAL net (ActionFormer-class) wants ~50?200+ labeled videos to beat heuristics; an Option-B VLM fine-tune is the sweet spot ? single-digit-thousands of clip?QA pairs (literature uses 13K?18K), which our ~128 rallies $\times$ augmentation $\times$ QA-task-multiplexing can approach without thousands of new matches, and RFT can

cold-start from hundreds. (d) The reusable format: one canonical JSONL `"rally record"` per (video_id, rally_ordinal) carrying [start,end], ending_reason, sport, source/provenance, split tag, + a pointer to the WASB trajectory slice; deterministic transcoders emit the heuristic-label view, the ActivityNet-TAL view, and the VLM clip?QA view, so EVERY label labeled once feeds all three model classes and all future generations.",

```
"options": [
  {
    "name": "(a) Format transcoders ? one corpus, three consumer formats",
    "what": "Treat the existing rally CSV rows
(rally_number,start_time,end_time,ending_reason,sport) as the source of truth and write
deterministic transcoders to each model class's expected format, instead of re-labeling per
consumer. View 1 (heuristic/LogReg distillation): we ALREADY do this ?
`backend/eval/training.py::label_candidates` runs the same IoU `match_intervals` to turn rally
intervals into positive/negative labels on WASB candidate windows; keep it. View 2 (TAL nets):
emit ActivityNet/ActionFormer JSON ? top-level `database`, per-video `{subset, duration, fps,
annotations:[{segment:[start,end] in SECONDS, label:'rally', label_id:0}]}` (verified against
ActionFormer's thumos14 loader). Our seconds-based [start,end] maps 1:1; we only add per-video
duration/fps (already in ffprobe_meta) and a train/val/test `subset`. View 3 (Option-B VLM):
emit clip?QA text pairs ? sample a window, ask 'List the [start,end] of every rally in this
clip' ? answer is the rally intervals normalized to clip-relative seconds; ending_reason yields
a SECOND free task ('why did rally N end?' ? winner/forced_error/...). One row, three formats,
zero re-labeling.",
    "pros": [
      "Zero new labeling: every consumer reads the same ~128 rallies; labels labeled once
feed heuristics + TAL + VLM and every future model generation",
      "Our schema is already 90% of the ActivityNet TAL schema ? `segment:[start,end]` in
seconds is exactly what ActionFormer/TriDet ingest; transcode is a thin map, not a re-
annotation",
      "ending_reason is a near-free second supervised task (rally-outcome QA) that
multiplies QA-pair yield from the SAME intervals",
      "Reuses the existing deterministic scorer (`metrics.match_intervals`) as both the
heuristic-labeler and the TAL/VLM eval grader ? one spine, no metric drift",
      "WASB trajectory CSV (Frame,Visibility,X,Y) attaches as an auxiliary per-frame feature
stream keyed by the same video_id+frame ? usable as TAL input features OR as a 'trajectory-
token' modality for Option B"
    ],
    "cons": [
      "VLM clip?QA generation needs a windowing/sampling policy (clip length, stride,
negative clips with no rally) ? a design choice that affects label balance",
      "ActivityNet-JSON path assumes you extract per-clip FEATURES (I3D/VideoMAE) for
classic TAL nets; that is extra preprocessing, though it's open-model and one-time",
      "Clip-relative vs absolute timestamps must be handled carefully in the VLM answer
format (Qwen2.5-VL aligns MROPE to absolute time ? pick one convention and lock it)"
    ],
    "cost_note": "Engineering-only; no labeling spend. A few hundred lines of transcoder +
tests; reuses existing metrics/annotations modules.",
    "license": "N/A (our own data + open formats). ActivityNet JSON schema and ActionFormer
loader are open/permissive references.",
    "privacy_note": "Transcoders run locally on licensed/owned corpus only; no user video,
no egress. Minors excluded at the corpus-membership layer before transcoding.",
    "fit_for_us": "strong"
  },
  {
    "name": "(b1) Self-training / teacher-student / pseudo-labeling ? open-teacher ONLY",
    "what": "Amplify the few labels by pseudo-labeling our ~26 GB unlabeled corpus with a
teacher that is OUR OWN model or an open MIT/Apache model (NEVER Gemini). Concretely: train a
first-pass rally model (heuristics+WASB, or a small TAL net) on the ~128 labeled rallies, run it
over unlabeled matches, keep only high-confidence pseudo-rallies (confidence-gated, à la
ALQA/PLR semi-supervised TAL), retrain on labeled+pseudo, iterate (noisy-student style with
temporal augmentation as the 'noise'). This is the canonical SS-TAL recipe; it directly attacks
the n?2 data starvation that currently makes the learned model lose to the baseline.",
    "pros": [
      "Stays inside guardrails: teacher is WASB/our-model/open-MIT ? paid-LLM outputs are
never a training signal",
      "SS-TAL with confidence-gated pseudo-labels (ALQA, PLR) is an established, published
```

```

recipe for exactly our 'few labeled + many unlabeled videos' regime",
    "Noisy-student framing (student ? teacher + augmentation noise) is a proven data-
efficiency multiplier (Xie et al. CVPR'20 lineage)",
    "Uses the corpus we already own; no per-rally vendor spend to grow training signal",
    "Pseudo-labels can bootstrap Option-B VLM SFT data (clip?QA pairs from confident
pseudo-rallies) without human labeling"
],
    "cons": [
        "Pseudo-label noise: low-quality pseudo-rallies degrade boundaries ? MUST confidence-
gate and ideally human-spot-check a sample (the PLR/ALQA papers exist precisely because naive
pseudo-labels hurt)",
        "Teacher blind spots propagate (WASB recall ceiling becomes the pseudo-label recall
ceiling) ? confirmed bottleneck in our own results",
        "Boundary precision from pseudo-labels is weaker than human [start,end]; treat pseudo
as recall-boosters, keep human labels as the boundary authority",
        "Iteration adds GPU cost on the single RTX 2070 Super (8 GB) ? feasible but slow;
cloud burst may be needed"
    ],
    "cost_note": "Mostly compute on owned GPU + occasional cloud burst. No new human
labeling required to START; small human audit budget to validate pseudo-label quality.",
    "license": "Teachers restricted to Apache-2.0/MIT/BSD (WASB=MIT, our own LogReg, open
TAL nets). Explicitly excludes Gemini/OpenAI/Anthropic outputs as training data.",
    "privacy_note": "All pseudo-labeling on owned/licensed corpus, local. Pseudo-labeled
clips inherit the same minors-excluded, training-opt-in constraints.",
    "fit_for_us": "strong"
},
{
    "name": "(b2) Temporal data augmentation ? multiply effective segments",
    "what": "Apply temporal augmentations that preserve rally semantics while changing the
signal: speed/time-warp ( $\pm 20\%30\%$ , rescale [start,end] accordingly), temporal jitter of
boundaries, VideoReverse (rally played backward still a rally), and segment-level
CutMix/MixUp/FadeMixUp to synthesize new rally/non-rally compositions. For VLM clip?QA, also
vary clip windows/stride so the same rally appears in many contexts. Scale-aware sampling
(randomize action timescale) is reported superior to spatial-only aug for video.",
    "pros": [
        "Multiplies effective labeled-segment count with ZERO new annotation ? directly
addresses tiny-n",
        "Speed/time-warp + scale-aware sampling are validated to improve temporally-
localizable features (RandAugment-T, FadeMixUp lineage)",
        "Boundary-preserving by construction (you transform the timestamps with the signal),
so it doesn't corrupt our hard-won [start,end] authority",
        "Cheap and deterministic; runs in the data loader for both TAL nets and VLM SFT",
        "Window/stride variation for clip?QA is a free QA-pair multiplier for Option B"
    ],
    "cons": [
        "Augmentation realism risk: extreme speed changes can break shuttle-physics cues WASB
relies on ? keep ranges moderate",
        "CutMix/MixUp across rallies can create unnatural boundaries; FadeMixUp (soft) is
safer than hard cut-paste for localization",
        "Doesn't add NEW information (same matches, same players, same court) ? complements
but cannot replace corpus diversity from new matches/venues",
        "Gains are incremental, not a substitute for the ~3?5 real matches the heuristic tuner
needs for stable cross-val"
    ],
    "cost_note": "Compute-only, in-loader; negligible marginal cost.",
    "license": "N/A ? transforms our own data.",
    "privacy_note": "Local, owned corpus; no egress.",
    "fit_for_us": "strong"
},
{
    "name": "(b3) Active learning ? label the most-informative match/segment next",
    "what": "Spend the Roora vendor budget where it buys the most: rank unlabeled
matches/clips by model uncertainty (boundary entropy, prediction disagreement between WASB-
heuristic and a TAL net, or high alignment-cost vs labeled prototypes) and send THOSE to Roora
first, rather than random/sequential matches. Active-learning-for-TAL papers (uncertainty/TPE

```

```

scoring, boundary-centric acquisition) show this beats random labeling under a fixed budget.",
  "pros": [
    "Maximizes IoU-lift-per-rupee on the constrained vendor budget ? exactly the solo-founder cost constraint",
    "Published SS-TAL+active-learning results show uncertainty-based selection beats random under fixed label budget",
    "Pairs naturally with (b1): pseudo-label everything, then human-label only the segments the model is LEAST sure about (the highest-value oracle queries)",
    "Boundary-centric acquisition targets our actual weak point (dStart/dEnd boundary error), not just presence/absence",
    "Disagreement signal is free ? we already run WASB-heuristic AND can run a TAL net; route their disagreements to humans"
  ],
  "cons": [
    "Needs a deployable uncertainty signal first; our LogReg gives probabilities but a richer TAL net gives better entropy estimates ? small upfront build",
    "Cold-start problem: with n?2 the model's uncertainty is itself unreliable; first 3?5 matches should still be diverse/random to bootstrap, THEN switch to active selection",
    "Risk of sampling bias (always querying hard cases) can skew the train distribution ? mix in some random/diverse picks",
    "Operationally couples the eval/training loop to the vendor pipeline (must re-rank between Roora batches)"
  ],
  "cost_note": "Saves vendor money net-net; small engineering cost for the scoring/ranking step. Highest ROI once n?~5.",
  "license": "N/A.",
  "privacy_note": "Selection runs locally; only the SELECTED owned/licensed clips go to the vendor through the existing single DPA-gated provider boundary.",
  "fit_for_us": "strong"
},
{
  "name": "(b4) Timestamp / point supervision ? cheaper labels for scale-out",
  "what": "For FUTURE labeling (new sports, ending_reason, scaling beyond Roora), prefer point/timestamp supervision (one click per rally + outcome) over dense [start,end] where boundary precision isn't the bottleneck. User study on Breakfast: timestamp supervision ? ? the annotation time of full supervision, SkipTag ? ?, and timestamp-supervised models reach near-full-supervision accuracy (e.g. 64.1% vs 68.0%). Point-level TAL is a mature sub-field.",
  "pros": [
    "Cuts annotation time substantially (timestamp ? ?, SkipTag ? ? of full) ? more matches per rupee",
    "Near-full-supervision accuracy is achievable from point labels with the right losses (HR-Pro, point-TAL methods)",
    "ending_reason is naturally point-like (one outcome per rally) ? fits timestamp supervision perfectly",
    "Lowers the barrier to expand to tennis/pickleball/table-tennis (collector's 5-sport shape) without dense re-labeling each sport",
    "Our existing dense [start,end] golden set becomes the high-quality 'boundary authority' that calibrates the cheaper point-labeled bulk"
  ],
  "cons": [
    "Point labels alone give weaker boundaries ? keep a dense-labeled golden CORE for eval and boundary calibration",
    "Requires point-supervision-aware training losses (not just feeding points to a full-supervision trainer)",
    "Mixing dense + point labels needs careful handling in the corpus format (label_type field) and the trainer",
    "Our current dense CSVs are already collected ? this is a forward-looking lever for the NEXT thousands of labels, not the existing 128"
  ],
  "cost_note": "Reduces future labeling cost ~33?66%. Existing dense labels unaffected.",
  "license": "N/A.",
  "privacy_note": "Same owned-corpus + vendor-DPA path; minors excluded.",
  "fit_for_us": "medium"
},
{

```

```

    "name": "(b5) Option-B specific: RL/RFT (GRPO) over SFT for label efficiency",
    "what": "When training the owned video-VLM (Qwen2.5-VL-7B / Qwen3-VL-8B / InternVL2.5,
all Apache-2.0/MIT), prefer reinforcement fine-tuning (GRPO/RFT) with a temporal-IoU reward over
pure SFT for the temporal-grounding objective. RFT is markedly more LABEL-efficient:
VideoChat-R1 reports +31.8 on temporal grounding with 'limited samples' and GRPO > SFT in their
ablation; VTG-R1 shows a quality-filtered 13K cold-start set beating an unfiltered 56K set. Our
reward = temporal IoU of predicted [start,end] vs our golden rally labels (the metric we already
compute).",
    "pros": [
        "RFT is data-efficient ? matches our exact situation: small, high-quality, hand-
verified label set is the RIGHT asset for RL, not a liability",
        "Reward = temporal IoU vs golden labels = our EXISTING deterministic scorer
(`metrics`) repurposed as the RL reward ? no new eval machinery",
        "Quality-over-quantity is empirically validated (filtered 13K > unfiltered 56K), so
our small clean set + augmentation can punch above its weight",
        "GRPO needs no separate reward model (uses verifiable IoU reward) ? operationally
simple, fits the nanochat-style one-command harness goal",
        "Bases are guardrail-clean (Qwen2.5-VL-7B/32B Apache-2.0; InternVL2.5 MIT; Gemma 4 if
Apache-2.0 confirmed)"
    ],
    "cons": [
        "RFT has higher COMPUTE cost than SFT (multiple rollouts per sample) ? heavy for one 8
GB RTX 2070 Super; likely needs cloud burst (RunPod/Lambda/Modal)",
        "Still wants a quality SFT cold-start first (cold-start data matters per VTG-R1) ? so
you need BOTH a clean SFT set AND the RL loop",
        "Reward hacking risk if IoU reward is naive (model emits over-long intervals) ? needs
precision/recall-balanced reward shaping",
        "VLM temporal grounding is still maturing; absolute accuracy on amateur single-camera
footage is unproven vs the broadcast-trained literature"
    ],
    "cost_note": "Compute-heavy (RL rollouts); plan cloud-GPU burst. But LABEL cost is low ?
the whole point. Benchmark self-hosted 7B vs Gemini per-call before committing (per
COMMERCIALIZATION C7).",
    "license": "Bases Apache-2.0/MIT only (Qwen2.5-VL-7B/32B, Qwen3-VL-4B/8B, InternVL2.5;
Gemma 4 pending license verify; Gemma 3 REJECTED). Reward = our own IoU, no paid-LLM signal.",
    "privacy_note": "Training data = owned/licensed corpus, training-opt-in only, minors
excluded. Final model runs locally (privacy lever).",
    "fit_for_us": "medium"
},
{
    "name": "(c) How many labeled matches per model class ? realistic bands",
    "what": "Heuristic threshold tuning (current WASB+windowing+gate): ~375 labeled matches
for STABLE leave-one-video-out cross-val; at n=2 the variance is too high, which is exactly why
our learned model loses to the baseline (this is a data-starvation artifact, not a method
failure). Small distilled classifier (our LogReg on trajectory features): also needs n~5
diverse matches before it reliably beats threshold tuning. From-scratch TAL net
(ActionFormer/TriDet class): wants ~50?200+ labeled videos to be competitive (these nets are
tuned on THUMOS14/ActivityNet-scale sets) ? NOT realistic for us soon, so deprioritize training
one from scratch. Option-B VLM fine-tune: the sweet spot ? single-digit-thousands of clip?QA
pairs (literature: 13K?18K), reachable from ~128 rallies x temporal augmentation x QA-task
multiplexing (rally-localization + ending_reason) x pseudo-labeled clips; RFT can cold-start
from HUNDREDS of high-quality pairs.",
    "pros": [
        "Sets honest, non-aspirational targets per model class ? prevents over-investing in a
from-scratch TAL net we can't feed",
        "Explains the current 'learned < baseline at n=2' result as expected data starvation,
not a dead end ? the fix is data, confirmed by our own docs",
        "Identifies the VLM path as the most label-EFFICIENT owned-model route given our asset
shape (few, clean labels + pretrained base)",
        "Gives a concrete corpus-growth milestone: get to ~5 matches to unblock honest
heuristic tuning NOW; build toward low-thousands QA pairs for Option B",
        "Aligns with the staged roadmap (heuristics ? Option A modular fusion ? Option B e2e)"
    ],
    "cons": [
        "The ~50?200 for from-scratch TAL is a literature-scale heuristic

```

```

(THUMOS/ActivityNet), not measured on our footage ? treat as order-of-magnitude",
    "Single-digit-thousands of QA pairs' for the VLM assumes augmentation + multiplexing
reach it; raw human rallies alone (128) are not enough without amplification",
    "Numbers are domain-transfer-sensitive: amateur single-camera footage may need MORE
labels than broadcast-trained literature implies",
    "Pose/audio-fused Option-A may shift these bands (more signal per label) ? revisit
after E1/pose probes"
  ],
  "cost_note": "Planning guidance; the actionable spend is getting from n?2 to n?5 matches
(small Roora batch) to unblock the heuristic loop immediately.",
  "license": "N/A.",
  "privacy_note": "N/A.",
  "fit_for_us": "strong"
},
{
  "name": "(d) Model-agnostic 'labeled-corpus' format ? the reusable spine",
  "what": "Define ONE canonical JSONL record per (video_id, rally_ordinal): {video_id
(file MD5), rally_ordinal, sport, start, end (seconds), ending_reason, source
(human|pseudo|vendor), annotator/provenance, label_type (segment|point), split (train|val|test,
assigned BY MATCH not by clip), trajectory_ref (pointer to the WASB CSV slice for this video)}.
Store it once (extends the existing collector export + candidate_segments lineage). Then
deterministic, tested transcoders emit: (1) the heuristic-label view (candidate-window pos/neg
via `match_intervals`), (2) the ActivityNet/ActionFormer TAL-JSON view (`database`?per-
video?`annotations:[{segment:[s,e],label,label_id}]`), (3) the VLM clip?QA view (rally-
localization + ending_reason tasks). Split is enforced by MATCH to prevent leakage
(CODE_MAP/PLATFORM 'split by match, not clip'). This is the asset the moat actually lives in.",
  "pros": [
    "Label once, consume everywhere and forever ? survives model-generation churn
(heuristics?TAL?VLM?whatever's next) because consumers read transcoder output, not raw labels",
    "Provenance + source + label_type fields make pseudo-labels, vendor labels, and human
labels first-class and auditable (critical for the no-paid-LLM + minors-excluded guardrails ?
each record carries who/what produced it and whether training-opt-in applies)",
    "split-by-match is baked into the format ? kills the data-leakage failure mode the
platform doc calls out (train|eval split by match not clip)",
    "Reuses existing infra: candidate_segments table is already the detector-agnostic
extension point; collector export already shares (video_id, ordinal, start, end) shape across 5
sports",
    "trajectory_ref keeps WASB as an attachable modality without bloating the rally record
? feeds TAL features AND Option-B multimodal tokens",
    "Same record powers eval (golden set) and training ? one corpus, graded by one scorer"
  ],
  "cons": [
    "Requires upfront schema discipline + migration of current CSVs into the JSONL (one-
time; the collector bridge/ADR-002 already points this way)",
    "Provenance/consent fields add bookkeeping the current flat CSV doesn't have ? but
they're non-negotiable given the guardrails",
    "Two timestamp conventions already exist in-repo (generic start,end vs golden
rally_number,start_time,end_time) ? the canonical record must absorb both and the transcoders
own the conversion",
    "trajectory_ref coupling means trajectory CSVs must be retained/versioned alongside
labels (storage, but cheap)"
  ],
  "cost_note": "One-time engineering: schema + migration + 3 transcoders + tests. No
labeling spend. Pays back on every future model.",
  "license": "Our own schema; emits open formats (ActivityNet JSON). No third-party
license entanglement.",
  "privacy_note": "Provenance/consent/minors-excluded/training-opt-in are FIELDS in the
record, enforced at corpus build; the format is the enforcement point for the legal
guardrails.",
  "fit_for_us": "strong"
}
],
"key_sources": [
  "https://arxiv.org/html/2407.07673v1",
  "https://pmc.ncbi.nlm.nih.gov/articles/PMC11798530/",

```

https://openaccess.thecvf.com/content/ICCV2023/papers/Xia_Learning_from_Noisy_Pseudo_Labels_for_Semi-Supervised_Temporal_Action_Localization_ICCV_2023_paper.pdf,
https://github.com/happyharrycn/actionformer_release/blob/main/README.md,
<https://arxiv.org/abs/2208.14856>,
https://joonyoung-cv.github.io/assets/paper/18_eccv_what_do.pdf,
<https://arxiv.org/html/2604.15173v1>,
<https://arxiv.org/pdf/2103.06669>,
https://www.ecva.net/papers/eccv_2022/papers_ECCV/papers/136640276.pdf,
<https://arxiv.org/abs/2504.06958>,
<https://arxiv.org/html/2507.18100v1>,
<https://arxiv.org/pdf/2502.13923>,
https://openaccess.thecvf.com/content_CVPR_2020/papers/Xie_Self-Training_With_Noisy_Student_Improves_ImageNet_Classification_CVPR_2020_paper.pdf,
<https://arxiv.org/abs/2211.04888>,
<https://ar5iv.labs.arxiv.org/html/2008.05721>,
<https://arxiv.org/pdf/2308.12608>

],
 "recommendation": "Stop treating the rally CSVs as eval-only files; promote them to a single model-agnostic \"labeled-corpus\" JSONL (d) ? one record per (video_id, rally_ordinal) carrying [start,end], ending_reason, sport, provenance/source, label_type, split-by-match, and a WASB-trajectory pointer ? with deterministic transcoders to three consumer formats (heuristic-window labels, ActivityNet/ActionFormer TAL JSON, VLM clip?QA pairs). This is the highest-leverage, lowest-cost move and it IS the moat. Then, in priority order: (1) IMMEDIATE ? grow from n?2 to n?5 matches via a small Roora batch; below ~5 matches the heuristic/LogReg tuner is data-starved and that, not the method, is why the learned model currently loses to the baseline. (2) NEAR-TERM, free ? temporal augmentation (moderate speed/time-warp, jitter, reverse, FadeMixUp) + open-teacher pseudo-labeling (WASB/our-model only, confidence-gated à la PLR/ALQA) over the owned ~26 GB corpus to multiply effective labels without violating the no-paid-LLM guardrail. (3) ONGOING ? route the Roora budget by active learning (uncertainty/disagreement/boundary entropy) once n?5, and adopt timestamp/point supervision for cheap scale-out to new sports and ending_reason. (4) DEFER but design-for-now ? Option B: fine-tune a guardrail-clean base (Qwen2.5-VL-7B Apache-2.0 / InternVL2.5 MIT) with a quality SFT cold-start THEN GRPO/RFT using temporal-IoU-vs-golden as the verifiable reward ? the data-efficiency literature (VideoChat-R1 +31.8 with limited samples; VTG-R1 filtered-13K > unfiltered-56K) says our small, clean, hand-verified set is exactly the right asset for RL, so we do NOT need thousands of new matches, we need amplification + a few clean ones. Do NOT invest in training a from-scratch TAL net (ActionFormer-class wants ~50?200+ labeled videos we won't have soon). Reuse `metrics.match\_intervals` as the single spine for heuristic-labeling, TAL eval, and the RL reward to avoid metric drift and data leakage.",

"confidence": "medium",

"caveats": [

"The 'how many matches' bands are calibrated from public TAL/VLM literature (THUMOS14/ActivityNet scale, VTG datasets of 13K?18K) and our own n=2 results, NOT measured on our amateur single-camera 60fps footage ? domain transfer from broadcast-trained literature is the dominant unknown and likely pushes our requirements UP (our own research doc flags this exact broadcast?amateur gap).",

"'Single-digit-thousands of clip?QA pairs' for Option B assumes augmentation + QA-task multiplexing + confident pseudo-labels reach it; ~128 raw human rallies alone are NOT enough ? amplification is load-bearing, not optional.",

"Pseudo-labeling inherits WASB's recall ceiling (our confirmed bottleneck): pseudo-rallies can only be as complete as the teacher, so they boost recall on what WASB sees and must be confidence-gated + human-spot-checked; treat human [start,end] as the boundary authority.",

"RFT/GRPO is LABEL-efficient but COMPUTE-heavy (multiple rollouts/sample) ? infeasible to run fully on one 8 GB RTX 2070 Super; budget cloud-GPU burst (RunPod/Lambda/Modal) and benchmark self-hosted-7B vs Gemini per-call before committing (COMMERCIALIZATION C7).",

"Timestamp-supervision savings (~? full time; SkipTag ~?; 64.1% vs 68.0% acc) come from the Breakfast action-SEGMENTATION user study, not racket-sports rally detection ? directionally relevant but not a measured number for our task.",

"Per the owner sequencing directive, this is a DESIGN blueprint: no Option-B implementation starts until P0?P4 scaffolding is solid and the golden-set corpus is healthy; the corpus-format (d) and n?5 (c) moves are the only items safe to act on now, and even those should get owner sign-off before code lands (CLAUDE.md: do not start implementation without explicit approval).",

"Base-model license must be re-verified per-checkpoint at build time (Qwen2.5-VL Apache-2.0 only at 7B/32B; 3B/72B custom; Gemma 4 Apache-2.0 status pending confirmation; Gemma

3 REJECTED) ? licenses change."

```
    ]
  },
  {
    "dimension": "COST + LICENSING + PRIVACY/COMPLIANCE for an owned, on-device-capable rally
temporal-segmentation model (and its Option-B end-to-end multimodal successor) for a Bangalore-
based solo-founder racket-sports highlight indexer.",
    "summary": "Three findings dominate. (1) COST: the two training paths are an order of
magnitude apart. A small TAL/segmentation head (ActionFormer-class) trains on ~4.5 GB VRAM, so
it fits the existing RTX 2070 Super (8 GB) locally at ~$0 marginal cost, or a few dollars of
rented GPU per full re-train; this matches today's bottleneck (golden data, not method). LoRA
fine-tuning a 2-7B video VLM needs ~16-24 GB VRAM ? it does NOT fit the 2070, so it is cloud-
only: realistic ranges are roughly $5-40 for a single LoRA run on rented GPU (RunPod A100 80GB
at ~$1.19-1.39/hr, ~4-30 GPU-hrs depending on clip count/epochs) or $2-30 on managed token-
priced backends (Together/Fireworks LoRA SFT ?16B at ~$0.48-0.50/M tokens). Full multimodal
fine-tuning is far heavier (~15 GPU-hrs/epoch on 4xA100 reported for a 7B VLM ? roughly
$70-300/epoch rented). (2) LICENSING: Apache-2.0-clean bases exist but the size matters
enormously for Qwen ? verified per-checkpoint: Qwen2.5-VL-7B = Apache-2.0 (clean), but 3B = Qwen
Research License (NON-commercial) and 72B = custom Qwen License (commercial-OK but 100M-MAU gate
+ 'Built with Qwen' attribution + competing-clause concerns). Qwen3-VL 4B/8B/30B/235B are
Apache-2.0. InternVL2.5 is MIT but some variants embed a Qwen-3B research-licensed component ?
verify the specific checkpoint. Gemma 3 is correctly REJECTED (custom Gemma Terms + Prohibited
Use Policy + viral distribution-flowdown). Gemma 4 reportedly ships Apache-2.0 but I could only
confirm this via secondary/editorial sources (not a primary Google page) and it is image-
multimodal, with no confirmed VIDEO capability ? treat as AMBER. Training on your OWN footage +
human rally labels is clean (you own the footage; labels are facts/timestamps, not copyrightable
expression; WASB code is MIT). (3) PRIVACY: Local inference is the strongest privacy lever
(video never leaves device ? no cross-border issue at all). For training/fallback backends:
Fireworks and Together both contractually state they do NOT train on customer data, both are
SOC2 Type II (Fireworks also HIPAA, BAAs), both offer Zero Data Retention; Google Vertex AI
states it 'won't use your data to train or fine-tune any models without your prior permission,'
with 24h in-memory-only caching and a ZDR path. India's DPDP Act 2023 is permissive on residency
(transfers allowed except to a govt 'negative list' ? currently empty as of Feb 2026), but
minors are under-16 and require verifiable parental consent ? reinforcing the committed 'exclude
minors from training pool' guardrail. The owned-model 'no paid-LLM outputs as training data'
rule is sound: Gemini/OpenAI/Anthropic terms restrict using outputs to build competing models,
so they remain eval/teacher-only.",
    "options": [
      {
        "name": "Path 1 ? Small TAL/segmentation head, trained LOCALLY (Option A, near-term)",
        "what": "ActionFormer-class or your distilled classifier on WASB-trajectory + (later)
audio/pose features ? rally [start,end]. ~4.5 GB training VRAM fits the RTX 2070 Super (8 GB) +
WSL2. No VLM, no per-call fees.",
        "pros": [
          "Effectively $0 marginal training cost on existing hardware; full re-train on a new
golden batch costs a few $ if you rent a GPU at all",
          "Fastest iteration loop ? re-train per labeled batch, CI-runnable, matches nanochat-
style operability goal",
          "All deps permissive (ActionFormer MIT, WASB code MIT, torch BSD) ? zero licensing
exposure",
          "Directly attacks today's real bottleneck (golden-data volume + WASB recall), not the
modeling method",
          "Cheapest at inference ? best steady-state margin per COMMERCIALIZATION C7"
        ],
        "cons": [
          "Inherits WASB candidate-window recall ceiling (the structural limit Option B is meant
to remove)",
          "Not end-to-end multimodal ? caps eventual quality vs a frames+audio+pose model",
          "Distilled learned model currently UNDERPERFORMS the threshold baseline at n=2 videos
? needs more data before it pays off"
        ],
        "cost_note": "~$0 on the local 2070 (8 GB) for the head; if rented, a full re-train is
single-digit dollars (e.g. <1-3 RunPod GPU-hrs at $0.34 RTX-4090 to $1.39 A100). Inference is
near-free.",
        "license": "MIT/BSD/Apache-2.0 across the stack ? clean.",
      }
    ]
  }
}
```

```

    "privacy_note": "Trains and infers fully local; user video never leaves device. No
cross-border/DPDP/GDPR transfer issue at all. Strongest privacy posture.",
    "fit_for_us": "strong"
  },
  {
    "name": "Path 2 ? LoRA fine-tune a 2-7B video VLM on RENTED cloud GPU (Option B bridge,
self-managed)",
    "what": "ms-swift/Unsloth LoRA/QLoRA on Qwen2.5-VL-7B (Apache-2.0) or Qwen3-VL-8B
(Apache-2.0) over 5k-20k clip?QA pairs, rented on RunPod/Lambda/Modal. ~16-24 GB VRAM ? A100
40/80GB class.",
    "pros": [
      "Removes WASB recall ceiling; moves toward the owned end-to-end multimodal target",
      "Per-run cost is modest and pay-as-you-go ? no standing GPU bill",
      "Apache-2.0 base (at 7B/8B) keeps fine-tuned weights private, gateable, monetizable",
      "RunPod is cheapest self-serve (A100 80GB ~$1.19-1.39/hr; H100 PCIe ~$1.99/hr; spot
40-50% off); you keep full control of weights + data"
    ],
    "cons": [
      "Does NOT fit the local 2070 (8 GB) ? must rent; adds a second stack to operate",
      "You own MLOps: data prep, leakage guards (split by match not clip), eval gate,
reproducibility",
      "Self-managed rental providers offer thin/no DPA vs managed APIs ? keep training data
to owned/consented footage only (which is already the plan)",
      "Needs 5k-20k curated QA pairs to be worth it; current 96+32 labeled rallies are far
below that band"
    ],
    "cost_note": "Single LoRA run ? $5-40 (?4-30 A100-hrs at ~$1.2-1.4/hr; 7B LoRA over
~1k-10k examples ? 0.5-5 hrs/epoch on one A100). Full multimodal FT is ~10-15x heavier
(~$70-300/epoch on 4xA100). Spot pricing cuts ~40-50%.",
    "license": "Qwen2.5-VL-7B = Apache-2.0 (VERIFIED clean). Qwen3-VL-8B = Apache-2.0. AVOID
Qwen2.5-VL-3B (Qwen Research License, non-commercial) and 72B (custom license, 100M-MAU gate +
attribution).",
    "privacy_note": "Training data = your owned/consented footage only; no user uploads, no
paid-LLM outputs. Self-managed rental ? verify the provider's own DPA/retention; prefer the
providers' ZDR/non-training defaults where offered. Inference can still be run locally after
training.",
    "fit_for_us": "medium"
  },
  {
    "name": "Path 3 ? Managed fine-tuning backend (Together AI or Fireworks AI)",
    "what": "Token-priced LoRA SFT on a supported open base, run as a managed job. You
upload the QA dataset; they return tuned weights/an endpoint.",
    "pros": [
      "Strong, explicit privacy contracts: Fireworks states it does NOT train on customer
content and is ZDR-by-default + SOC2 Type II + HIPAA/BAA; Together states no training without
opt-in, offers ZDR, SOC2 Type II, BAAs",
      "Cheap for the small band: LoRA SFT ?16B ~$0.48-0.50/M tokens ? a 5k-20k QA-pair run
is typically single-digit to low-tens of dollars",
      "Zero MLOps for the training run itself; fits the 'agent runs a fine-tune unattended'
operability goal",
      "Together fine-tuning catalog includes Qwen variants ? keeps you on an Apache-2.0
base"
    ],
    "cons": [
      "Less control over exact base checkpoint/version and weight portability than self-
hosting",
      "Per-run economics only beat rented GPU at low/medium volume; high sustained volume
favors owning the GPU",
      "Need to confirm the specific base offered is the Apache-2.0 size (e.g. Qwen 7B/8B,
not 3B/72B) before committing",
      "Data residency: confirm region ? these are US-centric; DPDP permits the transfer
today but Significant-Data-Fiduciary localization powers exist"
    ],
    "cost_note": "Together LoRA SFT: ?16B $0.48/M, 17-69B $1.50/M, 70-100B $2.90/M tokens
(full FT ~2.5x). Fireworks fine-tuning from ~$0.50/M (?16B). A realistic 5k-20k clip?QA-pair run

```

lands in the ~\$2-30 range.",

"license": "Base must be an Apache-2.0/MIT open model (Qwen 7B/8B, InternVL2.5 verified-clean variant). Provider tooling is fine; the weights' license is what governs monetization.",

"privacy_note": "Best contractual privacy of the cloud options: Fireworks 'does not use customer Content to train/improve models' + ZDR default; Together 'no training without opt-in' + ZDR setting. Still: route only owned/consented footage, never user uploads or paid-LLM outputs, and prefer in-region/ZDR config.",

"fit_for_us": "medium"

},

{

"name": "Path 4 ? Vertex AI managed tuning (the option the owner named)",

"what": "Google Vertex AI tuning/Model Garden for an open base or Gemini tuning. Managed, integrated with GCP storage/compute.",

"pros": [

"Clear data-governance statement: Google 'won't use your data to train or fine-tune any AI/ML models without your prior permission or instruction'; 24h in-memory-only caching; ZDR path available",

"Mumbai (asia-south1) region available ? can keep data in-India, easing DPDP/Significant-Data-Fiduciary concerns",

"Mature MLOps, integrates with the existing Gemini fallback already in the provider registry"

],

"cons": [

"GUARDRAIL TRAP: tuning Gemini itself does NOT yield an OWNED, on-device base ? it stays Google-served; only tuning an OPEN base in Model Garden gives portable weights",

"Tends to be pricier and more lock-in-prone than RunPod/Together for small runs",

"Open-base catalog/licensing on Vertex must be checked per-model; don't assume Apache-2.0",

"Heavier setup than a single rented GPU for a solo founder's first runs"

],

"cost_note": "GPU/tuning priced above the cheapest self-serve rentals; exact tuning cost depends on the base + tokens. Use only if in-India residency or GCP integration outweighs the premium.",

"license": "Acceptable ONLY if you tune an Apache-2.0/MIT open base with portable weights. Gemini-tuned models are not owned/on-device ? fails the owned-model goal (fine as a fallback, not as 'the owned model').",

"privacy_note": "Vertex paid tier: no training on customer data without permission; asia-south1 (Mumbai) keeps data in-India. Good privacy + residency story; weaker on the 'owned/on-device' requirement.",

"fit_for_us": "weak"

}

],

"key_sources": [

"https://www.runpod.io/pricing",

"https://lambda.ai/pricing",

"https://www.spheron.network/blog/gpu-cloud-pricing-comparison-2026/",

"https://www.together.ai/pricing",

"https://fireworks.ai/pricing",

"https://huggingface.co/Qwen/Qwen2.5-VL-7B-Instruct",

"https://huggingface.co/Qwen/Qwen2.5-VL-3B-Instruct/blob/main/LICENSE",

"https://huggingface.co/Qwen/Qwen2.5-VL-72B-Instruct/blob/main/LICENSE",

"https://github.com/QwenLM/Qwen3-VL/blob/main/LICENSE",

"https://huggingface.co/OpenGVLab/InternVL2_5-8B-MPO",

"https://github.com/OpenGVLab/InternVL/issues/900",

"https://ai.google.dev/gemma/terms",

"https://ai.google.dev/gemma/prohibited_use_policy",

"https://www.mindstudio.ai/blog/what-is-gemma-4-apache-2-license-commercial-ai-deployment",

"https://docs.cloud.google.com/vertex-ai/generative-ai/docs/data-governance",

"https://docs.fireworks.ai/guides/security_compliance/data_security",

"https://www.together.ai/blog/soc-2-compliance",

"https://www.meity.gov.in/static/uploads/2024/06/2bf1f0e9f04e6fb4f8fef35e82c42aa5.pdf",

"https://www.cyrilshroff.com/wp-content/uploads/2025/12/FAQs-DPDPA.pdf"

],

"recommendation": "Stage it. (1) NOW: keep building the small local TAL/segmentation head

(Path 1) ? it costs ~\$0 on the existing RTX 2070 Super, attacks the real bottleneck (golden-data volume + WASB recall), and carries zero licensing/privacy exposure. Spend the budget on golden labels (Roora/rally-annotator), not GPUs. (2) WHEN the corpus reaches the 5k-20k clip?QA-pair band: prototype the owned VLM with a LoRA fine-tune of Qwen2.5-VL-7B or Qwen3-VL-8B (both VERIFIED Apache-2.0) ? first on RunPod rented A100 80GB (Path 2, ~\$5-40/run, cheapest + full weight control) for the proof-of-concept; graduate to a managed backend (Path 3) if you want the stronger DPA + zero-MLOps and the per-run economics hold. Treat Vertex (Path 4) as in-India-residency or fallback infra, NOT as the owned model. (3) HARD licensing locks to encode in the C14 gate: APPROVE Qwen2.5-VL-7B/32B, Qwen3-VL-4B/8B/30B, InternVL2.5 (verified-clean variant); REJECT Qwen2.5-VL-3B (research/non-commercial), Qwen 72B (custom license), Gemma 3 (custom viral terms). Hold Gemma 4 as AMBER until you confirm Apache-2.0 from a primary Google source AND confirm video (not just image) capability. (4) PRIVACY: lean on local inference as the headline privacy lever; for any cloud training, enable ZDR/non-training configs (Fireworks/Together both contractually don't train on customer data; Vertex won't without permission), route ONLY owned/consented footage, keep minors out of the training pool (DPDP under-16 + verifiable parental consent), and keep paid-LLM outputs strictly eval/teacher-only.",

```
"confidence": "medium",
"caveats": [
  "GPU prices are volatile and provider-/region-/spot-dependent; the $ ranges (RunPod A100 80GB ~$1.19-1.39/hr, H100 PCIe ~$1.99/hr; Lambda H100 ~$3.29/hr) are 2026 snapshots ? re-quote before committing budget. RunPod adds ~$0.10-0.40/instance fees on top of GPU rates.",
  "Training-cost ranges depend heavily on dataset size, epochs, sequence/frame length, and LoRA vs full FT. The VLM LoRA figures assume the realistic 5k-20k clip?QA-pair band; full multimodal fine-tuning is roughly 10-15x heavier per epoch. Benchmark on your real workload before pricing the product (echoes COMMERCIALIZATION C7).",
  "Gemma 4 Apache-2.0 is the one license claim I could only confirm via secondary/editorial sources (mindstudio.ai), not a primary Google license page; and those sources describe it as image-multimodal with NO confirmed video capability. Verify both the license AND video support from ai.google.dev before relying on it as a base. Gemma 3 rejection is well-grounded (custom Gemma Terms + Prohibited Use Policy + distribution flow-down).",
  "InternVL2.5 is MIT, but a GitHub issue confirms some variants embed a Qwen-3B-Instruct component whose license can be the non-commercial Qwen Research License ? verify the EXACT checkpoint's bundled components before commercial use.",
  "Backend 'we don't train on your data' claims are from each vendor's own docs/blog (Fireworks, Together, Google) ? strong but self-asserted; get the signed DPA/BAA before sending any footage, and confirm the contracted region for India residency.",
  "India DPDP residency is permissive TODAY (empty negative list as of Feb 2026), but the govt retains Significant-Data-Fiduciary localization powers and the DPDP Rules are still settling ? re-check if you scale or get designated an SDF.",
  "The 'own footage + human labels = clean to train on' conclusion is a strong reading (you own the video; timestamps/reasons are non-copyrightable facts; WASB code is MIT) but is not formal legal advice ? the per-clip consent ledger + minors exclusion + separate training opt-in already planned are the right operational guardrails; confirm with counsel before scale."
]
```

2. assistant (2026-06-08T13:13:48.083Z)

I'll verify the load-bearing license and video-capability claims against primary sources. Let me start by loading the web tools and checking the most critical claims.

3. assistant (2026-06-08T13:13:51.631Z)

I'll verify each load-bearing license and video-capability claim against primary sources. Let me batch the most critical searches first, starting with the Qwen model licenses (the central claims).

4. user (2026-06-08T13:13:56.922Z)

Qwen2.5-VL-7B-Instruct: License and Capabilities

****License:**** Apache-2.0

****Video Support:**** Yes. The model demonstrates comprehensive video understanding capabilities, including:

- Processing videos exceeding 1 hour in length
- "Dynamic FPS sampling" for various video frame rates
- Temporal grounding with the ability to "pinpoint the relevant video segments"
- Support for video files, URLs, and frame sequences as input

The model card explicitly states it can "comprehend videos of over 1 hour" and features enhanced temporal reasoning through updated positional embeddings that enable it to "learn temporal sequence and speed."

5. user (2026-06-08T13:13:59.011Z)

License Analysis

****License Name:**** Qwen RESEARCH LICENSE AGREEMENT

****Key Restriction (Non-Commercial):****

The license explicitly states it grants rights "FOR NON-COMMERCIAL PURPOSES ONLY" and specifies: "If you are commercially using the Materials, you shall request a license from us."

****Direct Quote:****

"You are granted a non-exclusive, worldwide, non-transferable and royalty-free limited license...to use, reproduce, distribute, copy, create derivative works of, and make modifications to the Materials FOR NON-COMMERCIAL PURPOSES ONLY."

This is neither Apache-2.0 (which permits commercial use) nor a standard open-source license. It is Alibaba Cloud's proprietary research license with explicit non-commercial restrictions.

6. user (2026-06-08T13:13:59.932Z)

Qwen License Analysis

****License Name:**** Qwen LICENSE AGREEMENT

****MAU Threshold:**** Yes, the license contains a 100 million monthly active users threshold.

****Relevant Clauses:****

Commercial Use Restriction:

"If you are commercially using the Materials, and your product or service has more than 100 million monthly active users, you shall request a license from us."

Attribution Requirement:

"If you use the Materials or any outputs or results therefrom to create, train, fine-tune, or improve an AI model that is distributed or made available, you shall prominently display 'Built with Qwen' or 'Improved using Qwen' in the related product documentation."

****Summary:**** Users may freely use and modify the Qwen model, but those with commercial products exceeding 100 million monthly active users must obtain explicit authorization from Alibaba Cloud. Additionally, any derivative AI models must include prominent attribution crediting Qwen.

7. assistant (2026-06-08T13:14:02.154Z)

Three Qwen2.5-VL claims confirmed. Now the Qwen3-VL claims, Gemma 4 (the most uncertain/load-bearing reversal claim), and InternVL.

8. user (2026-06-08T13:14:06.650Z)

License Information

Based on the web page content provided, the exact license listed for the Qwen/Qwen3-VL-8B-Instruct model is:

****Apache-2.0****

This appears in the metadata section where it states "License: apache-2.0" alongside other model tags and specifications.

9. user (2026-06-08T13:14:07.328Z)

Qwen3-VL-4B-Instruct Model Details

****License:**** The model uses the "apache-2.0" license as indicated in the metadata tags.

****Video Input & Temporal Grounding:**** Yes, this model supports both capabilities. The documentation describes "Text?Timestamp Alignment" as a key architectural update that enables "precise, timestamp?grounded event localization for stronger video temporal modeling." Additionally, the model features "Long Context & Video Understanding" with "256K context, expandable to 1M; handles books and hours-long video with full recall."

10. user (2026-06-08T13:14:09.540Z)

InternVL3-8B Model Details

****License:**** The model operates under the Apache-2.0 License. As stated in the documentation: "This project uses the pre-trained Qwen2.5 as a component, which is licensed under the Apache-2.0 License."

****Underlying LLM Base:**** InternVL3-8B is constructed using Qwen2.5-7B as its language component. The architecture combines this with InternViT-300M-448px-V2_5 for the vision component.

****Video Understanding Support:**** Yes, the model supports video comprehension. The documentation includes dedicated sections demonstrating video multi-round conversation capabilities, with examples showing how to load video frames and process them through the model for tasks like describing video content.

11. assistant (2026-06-08T13:14:14.011Z)

Important finding on InternVL3-8B: the research finding claims "MIT on the InternVL3 wrapper." The HF card here describes the license as Apache-2.0 (from the Qwen2.5 component), not MIT. Let me verify the InternVL3 license tag precisely, and now tackle the most uncertain claim ? Gemma 4's existence and Apache-2.0 license. This is the load-bearing reversal claim.

12. user (2026-06-08T13:14:18.694Z)

Web search results for query: "InternVL3-8B license MIT OpenGVLab Hugging Face"

Links: [{"title": "OpenGVLab/InternVL3-8B-Pretrained · Hugging Face", "url": "https://huggingface.co/OpenGVLab/InternVL3-8B-Pretrained"}, {"title": "OpenGVLab/InternVL3-8B-Instruct · Hugging Face", "url": "https://huggingface.co/OpenGVLab/InternVL3-8B-Instruct"}, {"title": "OpenGVLab/InternVL3-8B · Hugging Face", "url": "https://huggingface.co/OpenGVLab/InternVL3-8B"}, {"title": "OpenGVLab/InternVL3-8B-hf · Hugging Face", "url": "https://huggingface.co/OpenGVLab/InternVL3-8B-hf"}, {"title": "OpenGVLab/InternVL3_5-8B-HF · Hugging Face", "url": "https://huggingface.co/OpenGVLab/InternVL3_5-8B-HF"}, {"title": "OpenGVLab/InternVL3_5-8B-Instruct · Hugging Face", "url": "https://huggingface.co/OpenGVLab/InternVL3_5-8B-Instruct"}, {"title": "ggml-org/InternVL3-8B-Instruct-GGUF · Hugging Face", "url": "https://huggingface.co/ggml-org/InternVL3-8B-Instruct-GGUF"}, {"title": "OpenGVLab/InternVL3_5-8B · Hugging Face", "url": "https://huggingface.co/OpenGVLab/InternVL3_5-8B"}, {"title": "OpenGVLab/InternVL3-8B at main", "url": "https://huggingface.co/OpenGVLab/InternVL3-8B/tree/main"}]

Based on the search results, I found information about the InternVL3-8B model from OpenGVLab on Hugging Face:

The project is released under the MIT License. Additionally, the project uses the pre-trained Qwen2.5 as a component, which is licensed under the Apache-2.0 License.

The model is available on Hugging Face under the OpenGVLab organization with several variants, including the pretrained and instruction-tuned versions at `OpenGVLab/InternVL3-8B`.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

13. user (2026-06-08T13:14:21.294Z)

Web search results for query: "Gemma 4 Google open model release Apache-2.0 license multimodal audio video"

Links: [{"title": "Google Opens Gemma 4 Under Apache 2.0 with Multimodal and Agentic Capabilities - InfoQ", "url": "https://www.infoq.com/news/2026/04/google-gemm4/"}, {"title": "What Is Google Gemma 4? The Apache 2.0 Open-Weight Model With Native Audio and Vision | MindStudio", "url": "https://www.mindstudio.ai/blog/what-is-google-gemma-4-apache-open-weight"}, {"title": "Google Releases Gemma 4 in Four Model Sizes Under Apache 2.0 License - gHacks Tech News", "url": "https://www.ghacks.net/2026/04/06/google-releases-gemma-4-in-four-model-sizes-under-apache-2-0-license/"}, {"title": "What Is Gemma 4? Google's Apache 2.0 Open-Weight Model With Native Audio and Vision | MindStudio", "url": "https://www.mindstudio.ai/blog/what-is-gemma-4-google-apache-open-weight-model"}, {"title": "What Is Gemma 4? Google's First Apache 2.0 Multimodal Model With Audio, Vision, and Function Calling | MindStudio", "url": "https://www.mindstudio.ai/blog/what-is-gemma-4-google-apache-2-multimodal-model"}, {"title": "Google Gemma 4 12B Brings Multimodal AI to 16GB Laptops, Free Under Apache 2.0", "url": "https://www.techtimes.com/articles/317758/20260604/google-gemma-4-12b-brings-multimodal-ai-16gb-laptops-free-under-apache-20.htm"}, {"title": "Google's new open source Gemma 4 12B analyzes audio, video ? and runs entirely locally on a typical 16GB enterprise laptop | VentureBeat", "url": "https://venturebeat.com/technology/googles-new-open-source-gemma-4-12b-analyzes-audio-video-and-runs-entirely-locally-on-a-typical-16gb-enterprise-laptop"}, {"title": "Google DeepMind Releases Gemma 4 12B: An Encoder-Free Multimodal Model with Native audio that runs on a 16 GB laptop - MarkTechPost", "url": "https://www.marktechpost.com/2026/06/03/google-deepmind-releases-gemma-4-12b-an-encoder-free-multimodal-model-with-native-audio-that-runs-on-a-16-gb-laptop/"}]

Based on the search results, here's what I found about Gemma 4:

Gemma 4 Release Overview

Google has announced the release of Gemma 4, a series of open-weight AI models, including variants with 2B, 4B, 26B, and 31B parameters, under the Apache 2.0 license.

Multimodal Capabilities

Key features include enhanced video and image processing, audio input on smaller models, and extended context windows up to 256K tokens. More specifically, all four Gemma 4 models come with native multimodal support for video and images at various resolutions, with OCR and chart understanding listed as key use cases. The E2B and E4B models also include native audio input for speech recognition.

Apache 2.0 License

Gemma 4 uses the Apache 2.0 license, which permits commercial use, modification, and redistribution without licensing fees. Gemma 4 now uses the Apache 2.0 license, offering standardized terms for commercial use, modification, and distribution without the need for a separate agreement with Google.

Latest 12B Model

Most recently, Google released Gemma 4 12B on June 3, 2026 ? a 12-billion-parameter open-weight model that processes text, images, audio, and video without separate encoder networks, and runs

on any laptop or workstation equipped with 16GB of RAM or VRAM. The model is available immediately, at no cost, under the Apache 2.0 license.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

14. assistant (2026-06-08T13:14:31.476Z)

InternVL3-8B is confirmed MIT (with Apache-2.0 Qwen2.5 component) ? the finding's "MIT wrapper + Apache base" framing is correct. Now the critical Gemma 4 verification. The research finding makes a specific claim: "Gemma 4 4B/12B (Apache-2.0)" with native video. But there's a discrepancy emerging ? one source lists sizes as "2B/4B/26B/31B" and another "4B/12B." Crucially, I must check whether this is from a PRIMARY Google source, not editorial. The fifth finding's own caveat flags this as AMBER. Let me check the official Google blog and the HuggingFace license file directly.

15. user (2026-06-08T13:14:37.505Z)

Web search results for query: "site:huggingface.co google Gemma 4 model card license"

Links: [{"title": "google/gemma-4-26B-A4B · Hugging Face", "url": "https://huggingface.co/google/gemma-4-26B-A4B"}, {"title": "google/gemma-4-E4B · Hugging Face", "url": "https://huggingface.co/google/gemma-4-E4B"}, {"title": "google/gemma-4-31B-it · Hugging Face", "url": "https://huggingface.co/google/gemma-4-31B-it"}, {"title": "google/gemma-4-E4B-it · Hugging Face", "url": "https://huggingface.co/google/gemma-4-E4B-it"}, {"title": "google/gemma-4-E2B · Hugging Face", "url": "https://huggingface.co/google/gemma-4-E2B"}, {"title": "google/gemma-4-31B · Hugging Face", "url": "https://huggingface.co/google/gemma-4-31B"}, {"title": "google/gemma-4-26B-A4B-it · Hugging Face", "url": "https://huggingface.co/google/gemma-4-26B-A4B-it"}, {"title": "google/gemma-4-E4B-it-qat-mobile-transformers · Hugging Face", "url": "https://huggingface.co/google/gemma-4-E4B-it-qat-mobile-transformers"}, {"title": "Welcome Gemma 4: Frontier multimodal intelligence on device", "url": "https://huggingface.co/blog/gemma4"}, {"title": "google/gemma-4-E2B-it · Hugging Face", "url": "https://huggingface.co/google/gemma-4-E2B-it"}]

Based on the search results, I found information about the Google Gemma 4 model licenses on Hugging Face.

The Google Gemma 4 models on Hugging Face are licensed under Apache 2.0. This applies across all variants of the Gemma 4 family, including the different model sizes (E2B, E4B, 26B A4B, and 31B) available on the platform.

The Gemma 4 family of multimodal models by Google DeepMind are truly open with Apache 2 licenses, making them freely available for use under the permissive Apache 2.0 open-source license terms.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

16. user (2026-06-08T13:14:38.147Z)

Analysis of Gemma 4 Blog Content

****Does the page exist and describe Gemma 4?***

Yes, this is an official Google blog post from April 2, 2026, announcing Gemma 4 as Google's newest open model family.

****License:****

The page states Gemma 4 is released under an ****Apache 2.0 license****, not Gemma Terms of Use. The exact language appears twice:

"breakthrough capabilities made widely accessible under an Apache 2.0 license"

and later: "Gemma 4 is released under a commercially permissive Apache 2.0 license."

****Model sizes listed:****

- Effective 2B (E2B)
- Effective 4B (E4B)
- 26B Mixture of Experts (MoE)
- 31B Dense

****Native video and audio capability:****

Yes. The page claims: "All models natively process video and images, supporting variable resolutions" and "the E2B and E4B models feature native audio input for speech recognition and understanding."

****License statement quote:****

"Gemma 4 is released under a commercially permissive Apache 2.0 license."

17. assistant (2026-06-08T13:14:42.851Z)

Gemma 4 is now confirmed from a PRIMARY Google source (blog.google) as Apache-2.0 with native video. This upgrades the finding's "AMBER" status. However, I notice an important nuance: the finding repeatedly cites "Gemma 4 4B/12B" sizes, but the actual sizes per Google are E2B/E4B/26B-A4B/31B ? and a separate June 3 release of a "12B." Let me verify the official HF model card license directly and check the 12B claim, then confirm Gemma 3 is still rejected (the guardrail). Let me also verify the most exploitable license traps: the ms-tcn Commons Clause and VideoMAE NC license.

18. user (2026-06-08T13:14:46.485Z)

Based on the webpage content provided, the license field states:

****"License: apache-2.0"****

The model card explicitly lists the license as Apache 2.0, with a link to the full Apache 2.0 license terms. There is no separate "Gemma" license or Gemma Terms of Use mentioned in the metadata shown on this page.

19. user (2026-06-08T13:14:48.220Z)

License Analysis

The file contains a ****MIT+CC License**** (MIT license combined with Commons Clause).

Yes, it does include a Commons Clause restriction that prohibits commercial use. The relevant restriction states:

"the grant of rights under the License will not include, and the License does not grant to you, the right to Sell the Software."

The license specifically defines "Sell" as practicing rights to provide products or services to third parties for a fee where the value derives substantially from the Software's functionality.

20. user (2026-06-08T13:14:48.333Z)

Based on the webpage content, the exact license listed for this VideoMAE model is ****cc-by-nc-4.0**** (Creative Commons Attribution-NonCommercial 4.0 International).

This is shown in the metadata section of the model card where it states "License: cc-by-nc-4.0". This is a non-commercial license that requires attribution.

21. assistant (2026-06-08T13:14:52.230Z)

Three more confirmed: Gemma 4 official HF card = apache-2.0; yabufarha/ms-tcn = Commons Clause (non-commercial trap confirmed); VideoMAE = cc-by-nc-4.0 (rejected, confirmed). Now let me verify the remaining TAL/TAS model licenses (the clean MIT claims), the VTimeLLM CC-BY-NC-ND claim, InternVideo2 licenses, and confirm Gemma 3 is still restrictive.

22. user (2026-06-08T13:14:55.598Z)

License Analysis

The file contains a **clean MIT License** with no additional restrictions.

The license grants permissions to "use, copy, modify, merge, publish, distribute, sublicense, and/or sell copies of the Software" without commercial restrictions. There is no Commons Clause or non-commercial limitation present in this license text.

Copyright holder: yabufarha (2019)

23. user (2026-06-08T13:14:56.231Z)

License Information for ASFormer

Based on the repository content provided, **ASFormer uses an MIT license**.

This is confirmed in two places:

1. The file listing shows a "LICENSE" file in the repository
2. The footer section explicitly states: "MIT license"

The MIT license is a permissive open-source license that allows free use, modification, and distribution of the software with minimal restrictions.

24. user (2026-06-08T13:14:58.689Z)

ActionFormer License and Input Flexibility

License: Yes, the ActionFormer repository uses an MIT license, as indicated in the repository's footer navigation and license section.

Input Feature Dimensions: The provided documentation does not contain information about whether the model accepts arbitrary input feature dimensions through an input projection layer. The materials discuss using the model with specific pre-extracted features (I3D 2048-d, TSP, SlowFast, etc.) but don't explain the underlying architecture's flexibility regarding input dimensions. You would need to examine the model code in `./libs/modeling`` to determine if such an input projection mechanism exists.

25. user (2026-06-08T13:14:59.936Z)

VTimeLLM License

Yes, the VTimeLLM repository uses the **Creative Commons Attribution-NonCommercial-NoDerivs 4.0 International License (CC-BY-NC-ND 4.0)**.

The repository explicitly states: "This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivs 4.0 International License." This means the software permits sharing with proper attribution but restricts commercial use and derivative works.

26. assistant (2026-06-08T13:15:04.138Z)

Note: the MS-TCN2 LICENSE shows copyright holder "yabufarha 2019" but it is a clean MIT (no Commons Clause) ? the finding's claim that sj-li/MS-TCN2 is clean MIT holds. ASFormer = MIT, ActionFormer = MIT, VTimeLLM = CC-BY-NC-ND 4.0 ? all confirmed. Now the remaining: TriDet,

DiffAct, InternVideo2 (MIT/Apache claims), TimeChat, and the Gemma 3 rejection guardrail.

27. user (2026-06-08T13:15:07.410Z)

License Information

License Type: MIT License

Copyright Holder: Dingfeng Shi (2023)

The file contains the standard MIT License text, which grants permission to use, modify, distribute, and sell the software with the conditions that the copyright notice and license disclaimer are included in all copies or substantial portions of the software.

28. user (2026-06-08T13:15:08.260Z)

License Type: MIT

Yes, the file contains an MIT License. The document begins with "MIT License" and is "Copyright (c) 2023 Daochang Liu."

The license grants permission to freely use, modify, distribute, and sublicense the software, provided that the copyright notice and license text are included in all copies. The software is provided without any warranty or liability.

29. user (2026-06-08T13:15:09.148Z)

Based on the web page content provided, the license listed for the InternVideo2-Stage2_6B model is **MIT**.

This is shown in the metadata section of the model card where it states "License: mit".

30. user (2026-06-08T13:15:10.363Z)

License Information for InternVideo2.5 Chat 8B

Based on the webpage content, the exact license listed in the metadata is **Apache-2.0**.

This is clearly indicated in the model information section where it states: "License: apache-2.0"

31. assistant (2026-06-08T13:15:13.847Z)

TriDet = MIT, DiffAct = MIT, InternVideo2-Stage2_6B = MIT, InternVideo2.5_Chat_8B = Apache-2.0 ? all confirmed. Now the final cluster: TimeChat (research/non-commercial claim), Momotor (no LICENSE/LLaMA claim), Gemma 3 rejection guardrail, and the VideoLLaMA3 non-commercial weights claim. Let me batch these.

32. user (2026-06-08T13:15:20.876Z)

TimeChat License

The repository uses **BSD-3-Clause license** (listed three times: LICENSE, LICENSE_Lavis.md, and LICENSE_Minigpt4.md).

Additionally, TimeChat includes a **research-only restriction**. The terms state: "Our TimeChat is just a research preview intended for non-commercial use only. You must **NOT** use our TimeChat for any illegal, harmful, violent, racist, or sexual purposes."

So while the underlying code follows the standard BSD-3-Clause open-source license, the project itself is explicitly restricted to non-commercial research applications only.

33. user (2026-06-08T13:15:21.048Z)

Gemma Terms of Use Summary

License Type

This is a **custom "Gemma Terms of Use"** agreement, NOT Apache 2.0. However, the page notes that "For Gemma 4 terms, see the Gemma 4 license" (which is Apache 2.0).

Scope

This agreement applies to **Gemma 1, 1.1, 2, 3, 3n, and variant models** listed in the Appendix. Gemma 4 is explicitly excluded and has separate licensing.

Key Restrictive Features

Distribution Requirements:

The terms require distributors to "include the use restrictions referenced in Section 3.2 as an enforceable provision in any agreement" governing downstream use. Recipients must receive "a copy of this Agreement." This creates mandatory flow-down obligations.

Use Restrictions:

Users cannot employ Gemma Services "for the restricted uses set forth in the Gemma Prohibited Use Policy" or "in violation of applicable laws and regulations."

Liability Disclaimer:

"THE GEMMA SERVICES, AND OUTPUTS, ARE PROVIDED ON AN 'AS IS' BASIS, WITHOUT WARRANTIES" of any kind, with Google disclaiming liability for damages.

Restrictive Nature:

This is moderately restrictive?it permits modification and distribution but enforces compliance requirements, prohibits certain use cases, and disclaims all warranties and liability.

34. user (2026-06-08T13:15:22.289Z)

Momentor Repository Licensing Analysis

LICENSE File

Based on the provided content, **no LICENSE file is explicitly mentioned** in the repository structure or documentation shown.

Underlying Model

Momentor is built on **LLaMA**, as evidenced in the acknowledgments section which lists "LLaMA" from Facebook Research among its foundational projects.

Stated License Information

The page content contains **no explicit license statement** for the Momentor project itself. The acknowledgments section credits several open-source projects including LLaMA, Vicuna, LLaVA, and others, but this does not constitute a stated license for Momentor.

Summary

The GitHub repository excerpt provided does not include visible licensing information, a LICENSE file reference, or explicit terms under which Momentor is distributed. Users would need to check the actual repository or contact the maintainers for clarification on the project's licensing terms.

35. user (2026-06-08T13:15:22.912Z)

VideoLLaMA3 License Summary

Code License: The VideoLLaMA3 code is released under the Apache 2.0 license.

****Model Weights Restriction:**** Yes, the model weights carry a non-commercial restriction. The documentation states: "The service is a research preview intended for ****non-commercial use ONLY****."

****Key Distinction:**** The Apache 2.0 code license permits commercial use, but the released model checkpoints themselves are restricted to research and non-commercial applications only. This creates a situation where developers can freely modify and commercialize the underlying code, but cannot commercialize the pre-trained weights without additional licensing.

36. assistant (2026-06-08T13:15:30.902Z)

Excellent ? this surfaces a meaningful correction. The finding claims TimeChat is "research-preview non-commercial" but says nothing about its actual code license being ****BSD-3-Clause****. More importantly, it lumps TimeChat with VTimeLLM/Momentor as "non-commercial" but the finding's TimeChat license line says "research-only/non-commercial" ? which is correct as to usability, though it omits that the underlying code is BSD. The substantive point (not usable as a base) holds.

All other claims confirmed: Gemma Terms of Use apply to Gemma 1/2/3/3n (Gemma 3 rejection valid, Gemma 4 explicitly excluded ? Apache); Momentor = no LICENSE + LLaMA-derived (confirmed); VideoLLaMA3 = Apache code / non-commercial weights (confirmed).

Now let me verify the key compute-platform privacy/pricing claims that are load-bearing for recommendations: Modal's no-access clause, RunPod pricing/Secure Cloud, Vertex no-train + India region, and Together's privacy gap. Let me batch the most decision-critical ones.

37. user (2026-06-08T13:15:35.137Z)

Modal's Data Privacy Claims

****No-Access Statement:****

Yes. Modal states: "Modal will never access or use: your source code. the inputs (function arguments) or outputs (function return values) to your Modal Functions. any data you store in Modal, such as in Images or Volumes."

****7-Day Deletion:****

Yes. For function inputs/outputs specifically, Modal confirms they are "deleted within a maximum TTL of 7 days."

****SOC 2 Type II:****

Yes. Modal states: "We have successfully completed a System and Organization Controls (SOC) 2 Type 2 audit."

38. user (2026-06-08T13:15:37.615Z)

Google Cloud Data Usage and Residency

No-Training Guarantee

Yes, Google explicitly states:

> ****"Customer data is never used by Google."**** We do not use customer data to train our models, or models used by others, without permission.

Data Residency for Asia

The article ****does NOT mention Asia-South1 (Mumbai) or India**** as a data residency option.

The only Asian regions listed for Generative AI on Vertex AI are:

> ****Asia: Japan, Singapore, and Korea****

For Vertex AI Search and Conversation, only multi-region options are provided (US and EU), with no India region mentioned.

Conclusion: While Google guarantees customer data won't be used for model training without permission, India/Mumbai (asia-south1) is notably absent from the announced data residency options for generative AI services as of November 2023.

39. user (2026-06-08T13:15:38.796Z)

RunPod GPU Pricing

Based on the pricing page content, here are the hourly rates for these GPUs on **Pods** (the pricing shown applies to Community Cloud):

A100 80GB Options:

- A100 PCIe: "\$1.39/hr"
- A100 SXM: "\$1.49/hr"

H100 Options:

- H100 PCIe: "\$2.89/hr"
- H100 SXM: "\$3.29/hr"
- H100 NVL: "\$3.19/hr"

Regarding pricing tiers: The page displays two column headers labeled "Community Cloud" and "Secure Cloud" for the Pods section, but the content provided only shows specific pricing figures without differentiating which tier each price belongs to. The exact dollar amounts listed above represent the rates visible in the pricing table, though the distinction between the two cloud options is not clearly elaborated in this extract.

40. assistant (2026-06-08T13:15:45.180Z)

Key finding on the Vertex India-region claim: the cited gen-AI data-residency blog explicitly lists Asia residency as Japan/Singapore/Korea and does NOT include asia-south1/Mumbai for generative AI services. The findings acknowledge this nuance ? that asia-south1 GPUs exist for Compute Engine custom training but are excluded from the managed gen-AI residency list. This is internally handled in their caveats. The Modal no-access and Google no-train claims are confirmed verbatim.

On RunPod pricing: the page shows A100 80GB PCIe at \$1.39/hr (the findings cited "\$1.19-\$1.29/hr Secure"). Current pricing is slightly higher than the finding's low end. Let me verify the Vertex AI India region claim more directly (it's load-bearing for the "only strong-privacy option with India region" recommendation) and the Together AI no-train gap.

41. user (2026-06-08T13:15:52.522Z)

Web search results for query: "Google Cloud Compute Engine A100 H100 GPU availability asia-south1 Mumbai region 2026"

Links: [{"title": "GPU locations | Compute Engine | Google Cloud Documentation", "url": "https://docs.cloud.google.com/compute/docs/regions-zones/gpu-regions-zones"}, {"title": "GPU machine types | Compute Engine | Google Cloud Documentation", "url": "https://docs.cloud.google.com/compute/docs/gpus"}, {"title": "Holori - List of GPU available in the different GCP Zones", "url": "https://holori.com/list-of-gpu-available-in-the-different-gcp-zones/"}, {"title": "Available GPUs | Cloud Workstations | Google Cloud Documentation", "url": "https://docs.cloud.google.com/workstations/docs/available-gpus"}, {"title": "GPU support for services | Cloud Run | Google Cloud Documentation", "url": "https://docs.cloud.google.com/run/docs/configuring/services/gpu"}, {"title": "Machine Types with GPU in Google Cloud Platform", "url": "https://gcloud-compute.com/gpu.html"}, {"title": "Overview of creating an instance with attached GPUs | Compute Engine | Google Cloud Documentation", "url": "https://docs.cloud.google.com/compute/docs/gpus/create-vm-with-gpus"}, {"title": "Cloud GPU Pricing Comparison India [2026]: A100, H200 & More", "url": "https://acecloud.ai/blog/cloud-gpu-pricing-comparison/"}, {"title": "VM instance pricing | Google Cloud", "url": "https://cloud.google.com/compute/gpus-pricing"}, {"title": "GitHub - doitintl/gpu-finder · GitHub", "url": "https://github.com/doitintl/gpu-finder"}]

Based on the search results, here's what I found regarding GPU availability in the asia-south1 (Mumbai) region:

Summary

The search results don't provide specific information about A100 and H100 GPU availability in the asia-south1 (Mumbai) region for 2026.

However, the official Google Cloud documentation reveals some relevant details:

****H100 Availability:****

For A3 High (NVIDIA H100) machine types, the following regions and zones are available with limited capacity. To access these regions and zones, contact your account team. The documentation only lists St. Ghislain, Belgium (europe-west1-b) as an example location, but doesn't mention asia-south1.

****A100 Availability:****

For A2 Standard (NVIDIA A100 40GB) machine types, the a2-megagpu-16g machine is only available in the following regions and zones, but the specific regions aren't fully detailed in the search results.

****Mumbai Region Note:****

Regarding the asia-south1 region itself, asia-south1 (Mumbai) is available by invitation only.

****Recommendation:****

For current information on GPU availability in the asia-south1 region, I recommend checking the [official Google Cloud GPU regions and zones documentation](<https://docs.cloud.google.com/compute/docs/regions-zones/gpu-regions-zones>) directly or contacting your Google Account team, as these details are subject to change and may require special access arrangements.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

42. user (2026-06-08T13:15:52.915Z)

Privacy and Data Ownership Claims

Based on the provided webpage content, here's what Together AI explicitly states:

Data Training

The page does **not** contain language stating "does NOT train on customer data."

Privacy/Compliance

Together AI mentions:

- ****SOC 2 Type II****: "AICPA SOC 2 Type II" ?
- ****Data ownership****: "Your data and models remain fully under your ownership, safeguarded by robust security measures."
- ****Security features****: "Encryption in transit and at rest. Deploy storage in regions matching your data residency requirements"

However, the page does not explicitly mention:

- A Data Processing Agreement (DPA)
- Zero data retention policy
- Explicit language about downloading fine-tuned model weights

Model Ownership

The page states customers retain ownership but doesn't specifically address downloadability of weights. One customer testimonial mentions "LoRA serving" capabilities, suggesting models remain under customer control within the platform, but the content doesn't explicitly confirm local weight downloads are available.

****Notable gap**:** The security statement emphasizes ownership and encryption but stops short of explicitly stating "we do not train on your data" or "zero retention after training completion."

43. assistant (2026-06-08T13:16:01.132Z)

Both confirmed: Together AI's page does NOT contain an explicit no-train clause (the finding's "BLOCKED until DPA" caution is correct), and the asia-south1 GPU availability is "by invitation only" / contact-account-team ? which supports the finding's caveat that asia-south1 capacity must be confirmed and is capacity-constrained, but also weakens "genuine India residency-at-rest, unique among strong options" as a clean ready-to-use claim.

I have now verified all the load-bearing license claims (the primary focus) plus the key privacy/pricing claims. Let me do one final check on the two remaining specific license sub-claims that gate recommendations: InternVL3-78B (Qwen2.5-72B MAU cap inheritance) and Qwen3-VL larger sizes, since the findings make per-size approval/rejection calls.

44. user (2026-06-08T13:16:07.154Z)

Web search results for query: "InternVL3 license sizes 78B Qwen2.5-72B 38B 9B InternLM model card commercial use"

Links: [{"title": "OpenGVLab/InternVL3-78B · Hugging Face", "url": "https://huggingface.co/OpenGVLab/InternVL3-78B"}, {"title": "GitHub - OpenGVLab/InternVL: [CVPR 2024 Oral] InternVL Family: A Pioneering Open-Source Alternative to GPT-4o. ??GPT-4o????????????? · GitHub", "url": "https://github.com/OpenGVLab/InternVL"}, {"title": "OpenGVLab/InternVL2_5-78B · Hugging Face", "url": "https://huggingface.co/OpenGVLab/InternVL2_5-78B"}, {"title": "OpenGVLab/InternVL3-38B · Hugging Face", "url": "https://huggingface.co/OpenGVLab/InternVL3-38B"}, {"title": "OpenGVLab/InternVL3-9B · Hugging Face", "url": "https://huggingface.co/OpenGVLab/InternVL3-9B"}, {"title": "InternVL3", "url": "https://internvl.github.io/blog/2025-04-11-InternVL-3.0/"}, {"title": "OpenGVLab/InternVL3-78B-Instruct · Hugging Face", "url": "https://huggingface.co/OpenGVLab/InternVL3-78B-Instruct"}, {"title": "Quick Start of InternVL 3.0 Series ? InternVL", "url": "https://internvl.readthedocs.io/en/latest/internvl3.0/quick_start.html"}, {"title": "InternVL3-78B | PromptLayer Models", "url": "https://www.promptlayer.com/models/internvl3-78b/"}]

Based on the search results, here's the information about InternVL3 model licenses and sizes:

License Information

InternVL3-78B is released under the MIT License and uses pre-trained Qwen2.5 as a component, which is licensed under the Qwen License.

InternVL3-38B is released under the MIT License and uses pre-trained Qwen2.5 as a component, which is licensed under the Apache-2.0 License.

For the InternVL3-9B model, the search results indicate it uses InternLM 3 as the language component.

Model Sizes and Architecture

The InternVL3 series includes models of the following sizes: InternVL3-9B (using InternLM3-8B-instruct), InternVL3-14B (using Qwen2.5-14B), InternVL3-38B (using Qwen2.5-32B), and InternVL3-78B (using Qwen2.5-72B).

All InternVL3 models use the same vision architecture but are paired with different language model components based on their size.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

45. assistant (2026-06-08T13:16:11.816Z)

This confirms the per-size InternVL3 nuance precisely: InternVL3-78B = MIT wrapper but Qwen2.5-72B component under the Qwen License (the 100M-MAU-gated one), while InternVL3-38B = MIT + Apache-2.0 Qwen2.5-32B. The finding's per-size analysis ("REJECT InternVL3-78B; ?38B clean") is accurate.

I have now verified every load-bearing license claim against primary sources plus the most decision-critical privacy/pricing claims. Let me compile the structured verdict.

46. user (2026-06-08T13:17:28.467Z)

Structured output provided successfully

47. assistant (2026-06-08T13:17:39.852Z)

I have completed the adversarial verification. All 12 distinct license claims and the key video-capability and privacy/pricing claims were checked against primary sources (HuggingFace model cards, GitHub LICENSE files, official Google/Modal/Together/RunPod pages).

Bottom line: the license findings are highly accurate. Every per-size license call (Qwen2.5-VL 7B Apache / 3B Research-NC / 72B 100M-MAU; Qwen3-VL 4B/8B Apache; InternVL3 MIT+Apache with 78B inheriting the Qwen-72B cap; the ms-tcn Commons-Clause trap; VideoMAE NC; VTimeLLM CC-BY-NC-ND; Momentor no-LICENSE; all the MIT TAL/TAS repos) held up against primary sources.

The most consequential change is on Gemma 4: I upgraded it from finding #5's conservative AMBER to verified GREEN. Primary sources (the official Google blog plus the official google/gemma-4-E4B HF card) confirm both the Apache-2.0 license AND native video ? resolving finding #5's stated doubt that video was unconfirmed. The only imprecision is size-naming (official family is E2B/E4B/26B-A4B/31B, not the "4B/12B" shorthand the findings use).

Two recommendation-relevant softenings, neither of which flips a verdict: (1) the Vertex "genuine India residency-at-rest, unique" claim is capacity-gated ? asia-south1 high-end GPU access appears invitation-only, so it's conditional not turnkey; (2) RunPod \$/hr figures are stale on the low side (live A100 80GB \$1.39/hr vs cited \$1.19? \$1.29), already covered by the findings' own ±15?25% volatility caveat. One claim ? that ActionFormer/ASFormer ingest arbitrary low-dim WASB features via input projection ? I marked uncertain because it's a code-architecture claim not confirmable from the README; it's not a license issue and the findings already flag it as unproven.